Universal Shape Replicators via Self-Assembly with Attractive and Repulsive Forces

Symposium on Discrete Algorithms

January 16, 2017

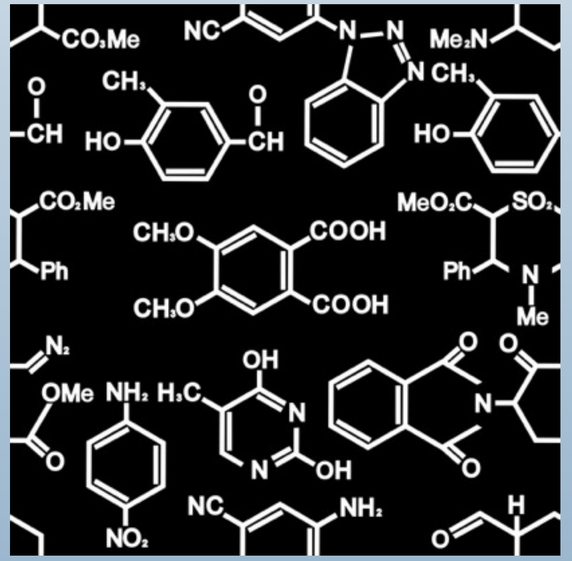
Cameron Chalk[1], Eric D. Demaine[2], Martin L. Demaine[2], Eric Martinez[1],
Luis Vega[1], Robert Schweller[1], Tim Wylie[1]

[1]University of Texas – Rio Grande Valley

[2]Massachusetts Institute of Technology

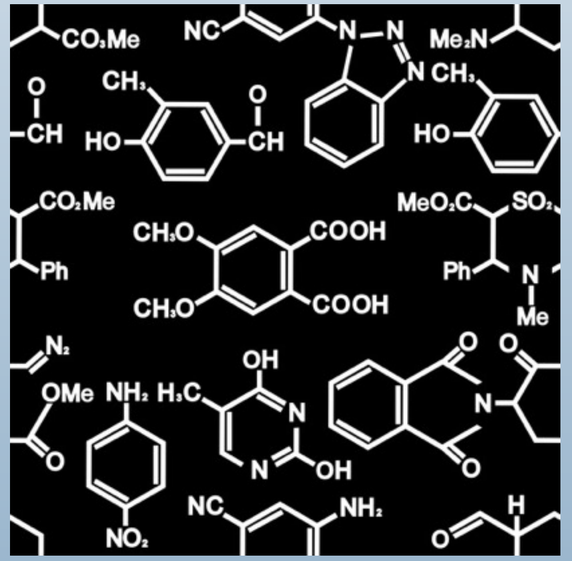
Self-Assembly

Chemistry



Self-Assembly

Chemistry

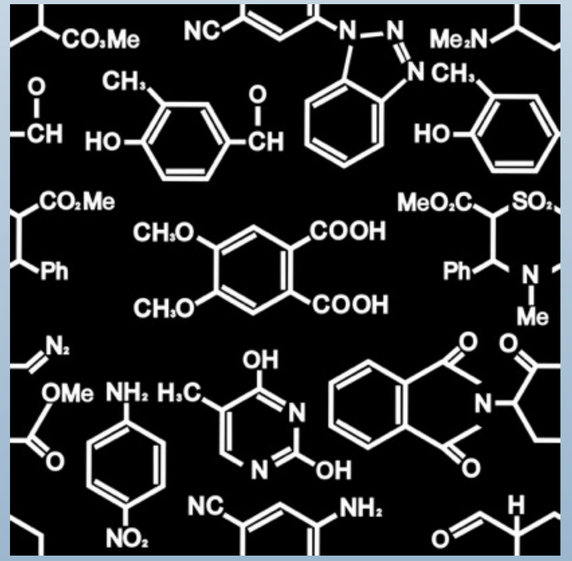


Sociology



Self-Assembly

Chemistry



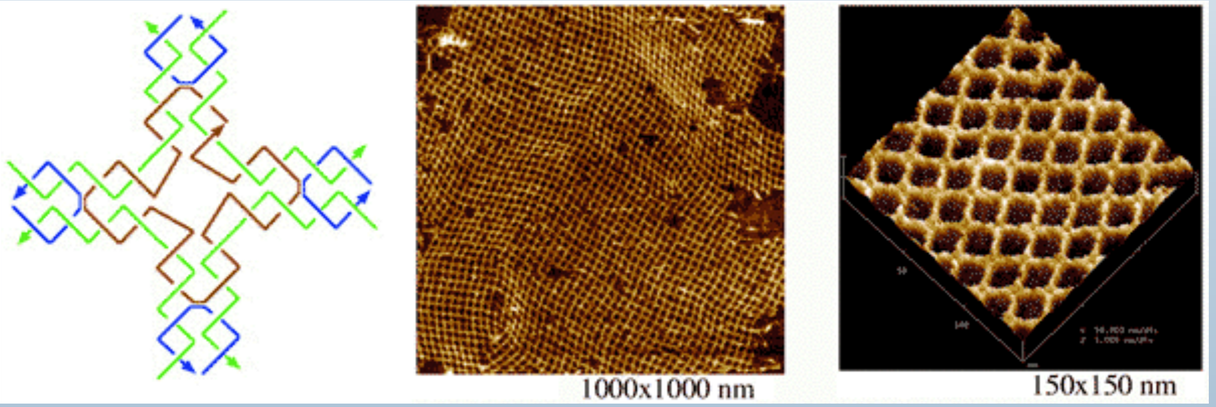
Biology



Sociology

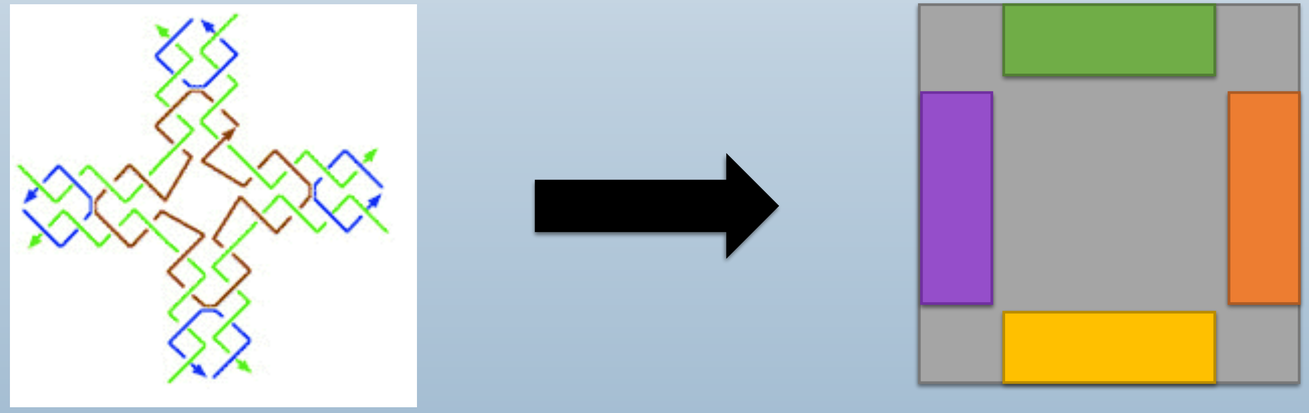


Self-Assembly



- Yan, H., Park, S.H., Ginkelstein, G., Reif, J.H. & LaBean, T.H.

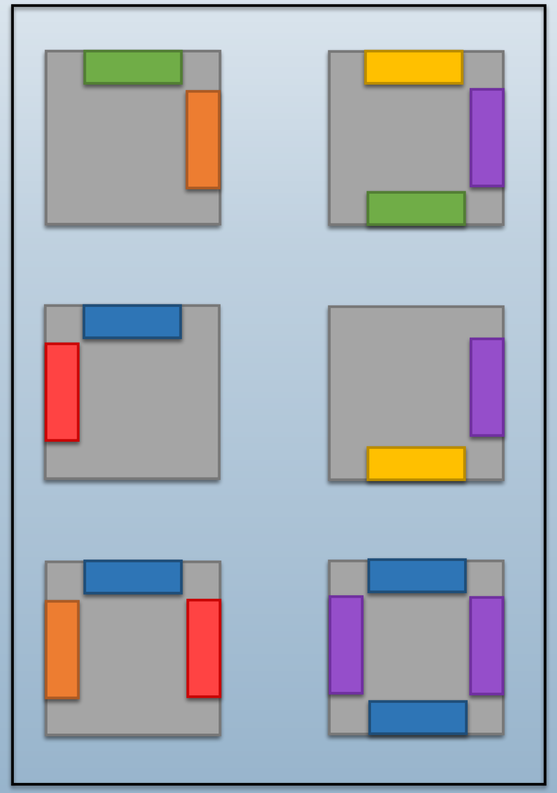
Tile Self-Assembly



(Winfree, 1998)

Tile Self-Assembly
(Rothemund, Winfree, Adleman)

Tileset:



Glues:

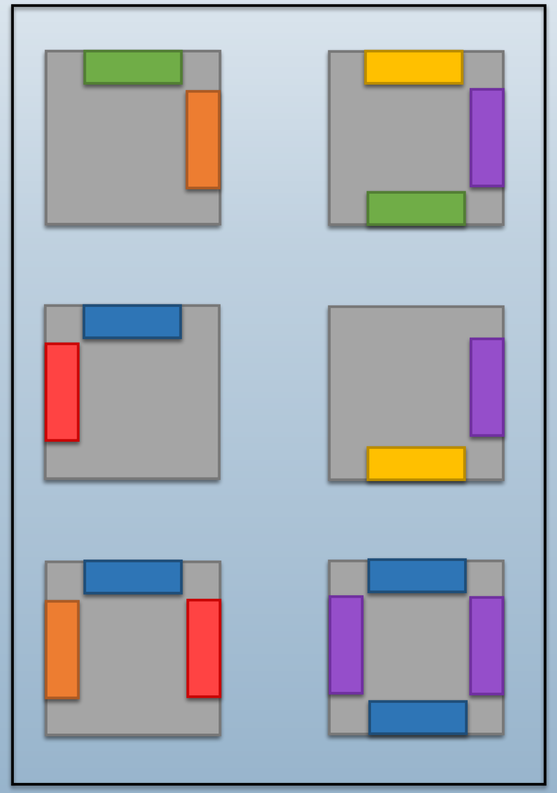
G(g) = 2
G(o) = 2
G(y) = 2
G(r) = 2
G(b) = 1
G(p) = 1

Temp. = 2

Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2
G(o) = 2
G(y) = 2
G(r) = 2
G(b) = 1
G(p) = 1

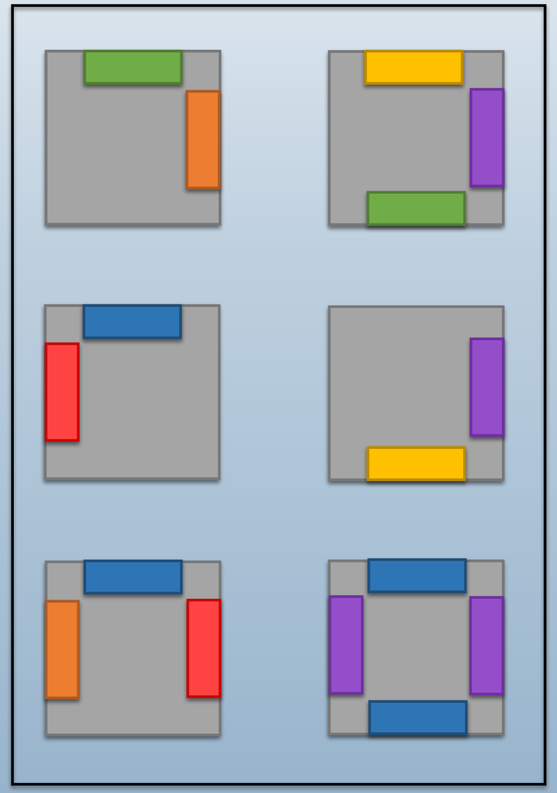
Temp. = 2

Producible Assemblies:

Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

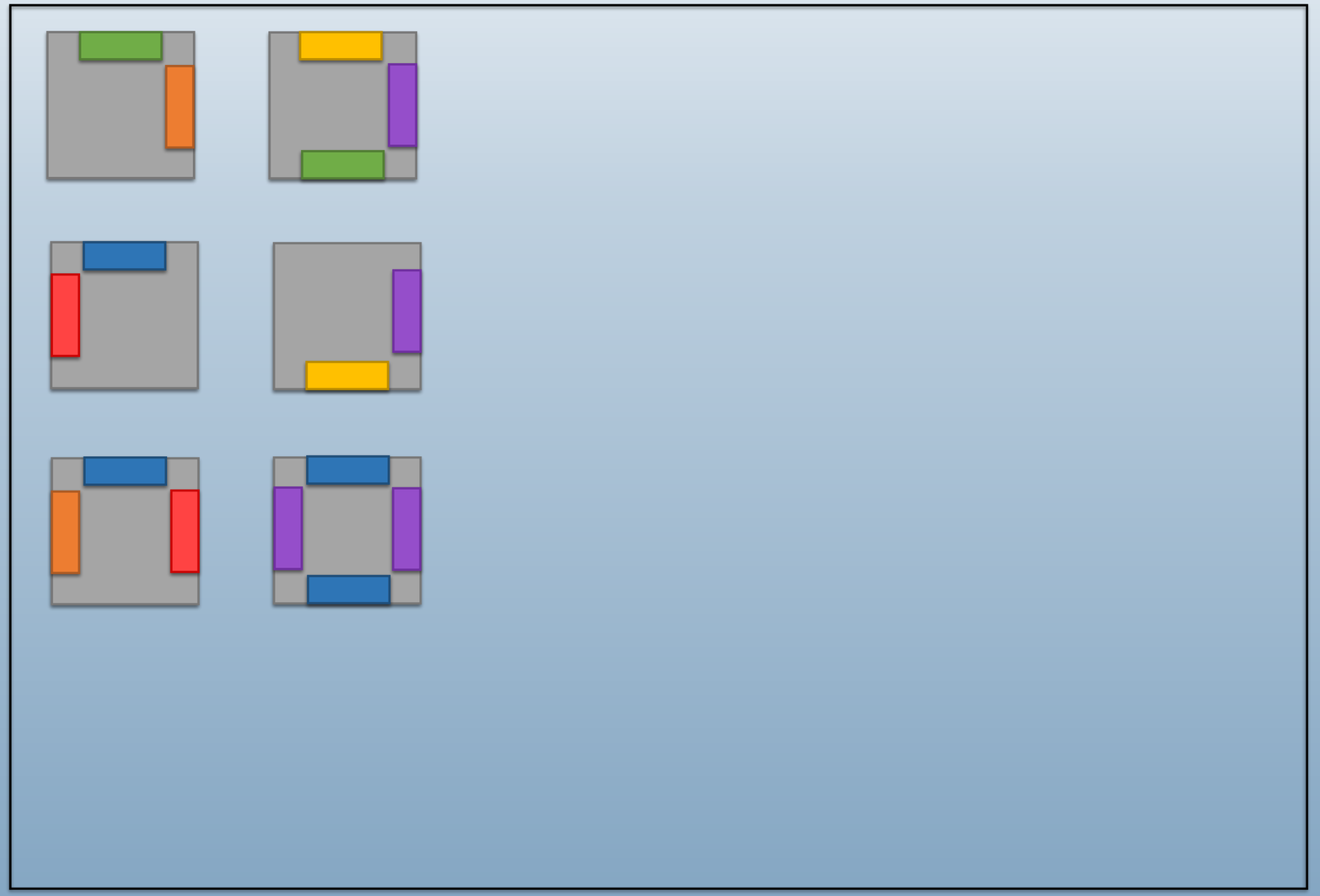
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

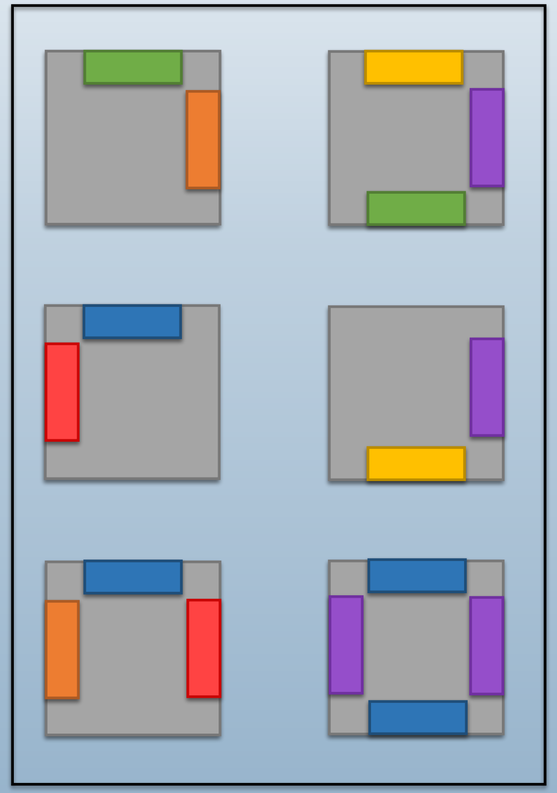
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

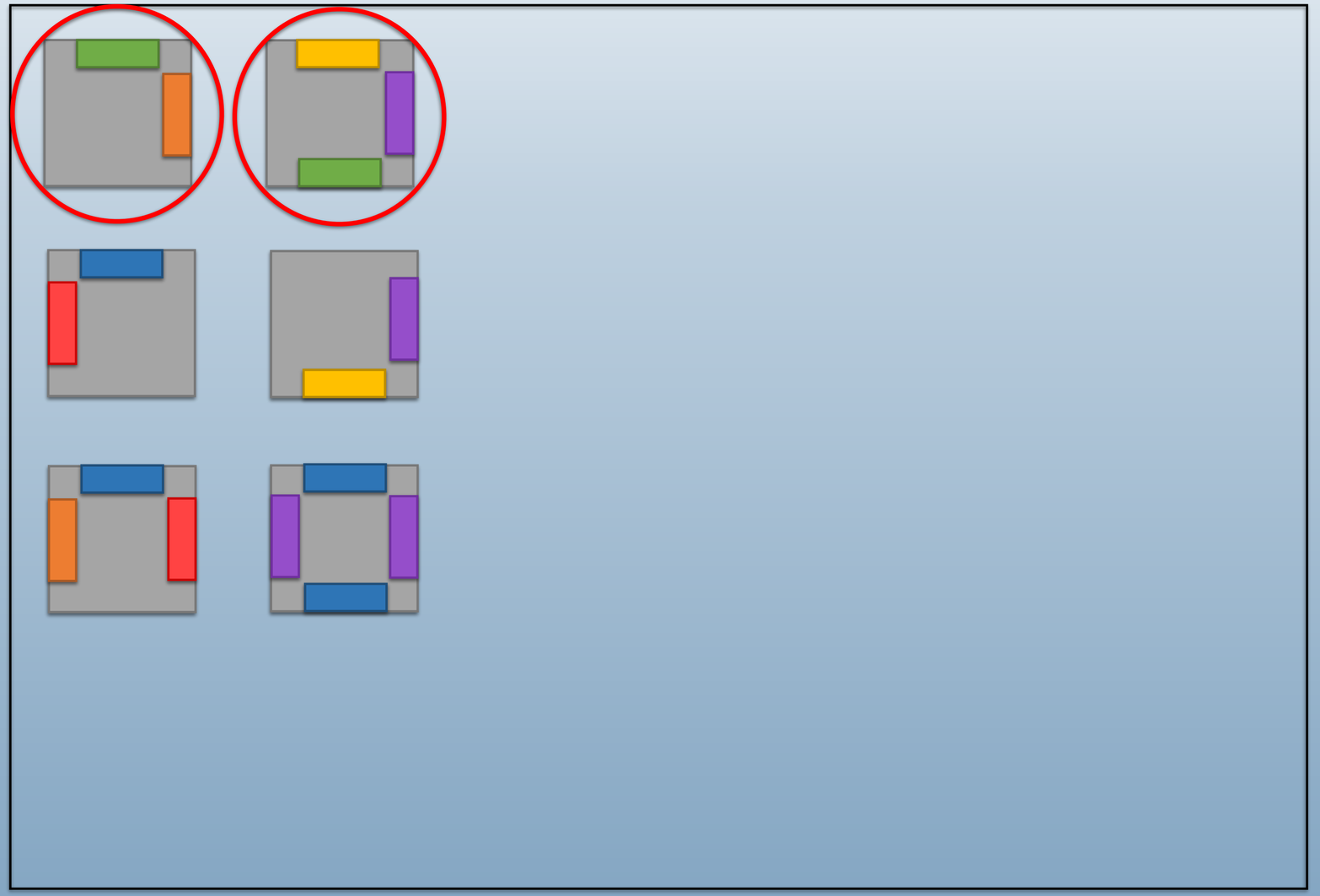
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

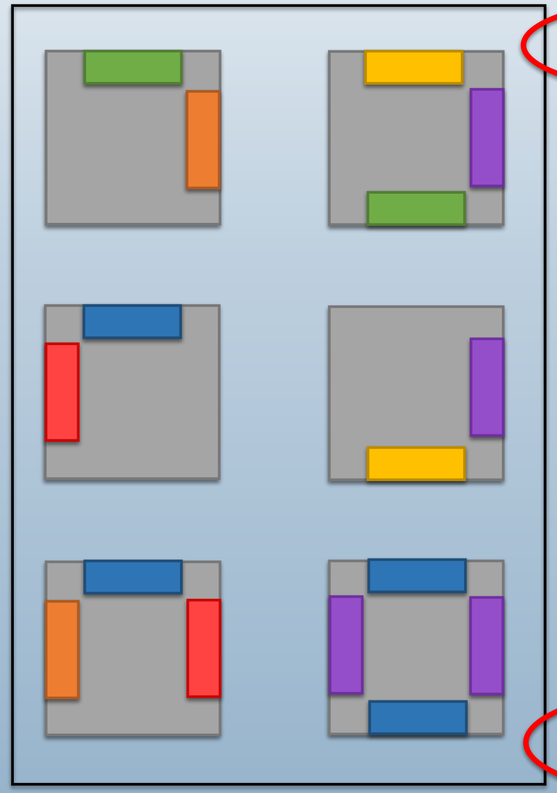
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

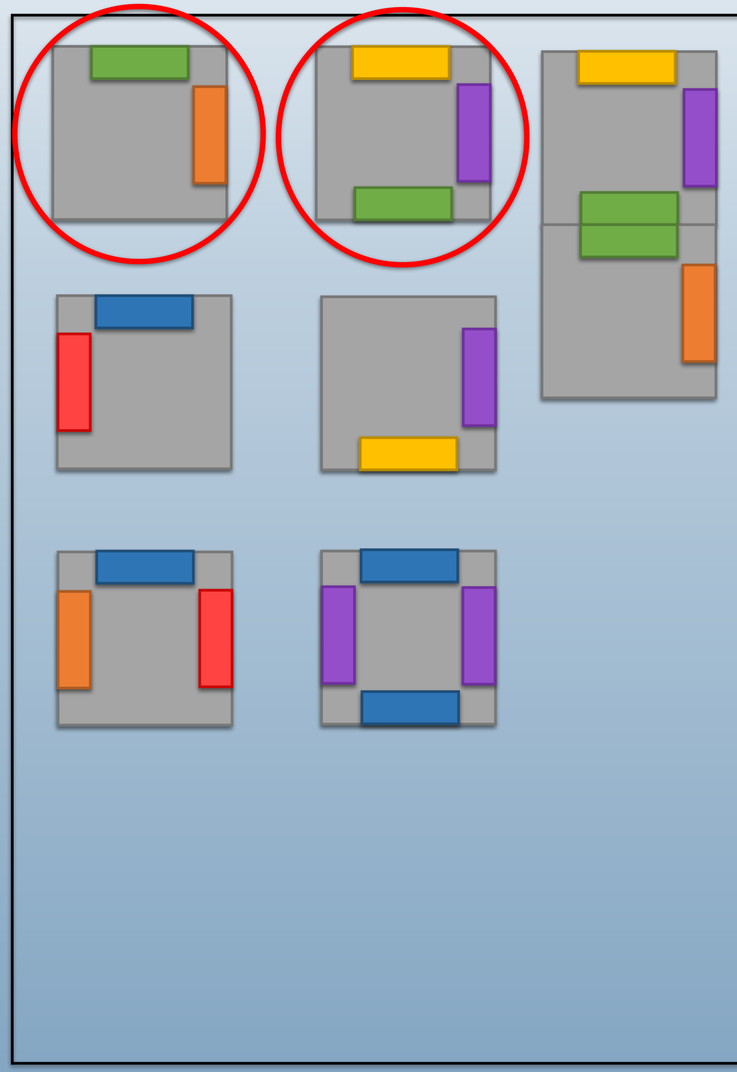
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

Producible Assemblies:



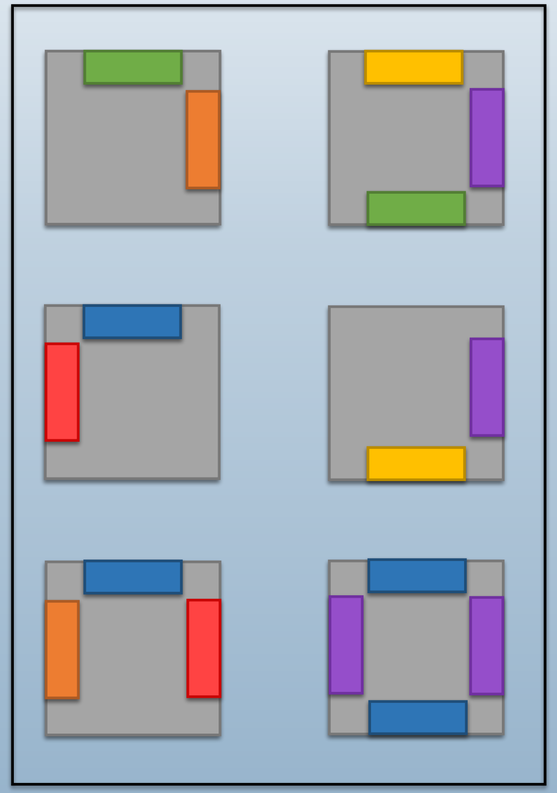
G(g) ≥ Temp.

is producible

Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

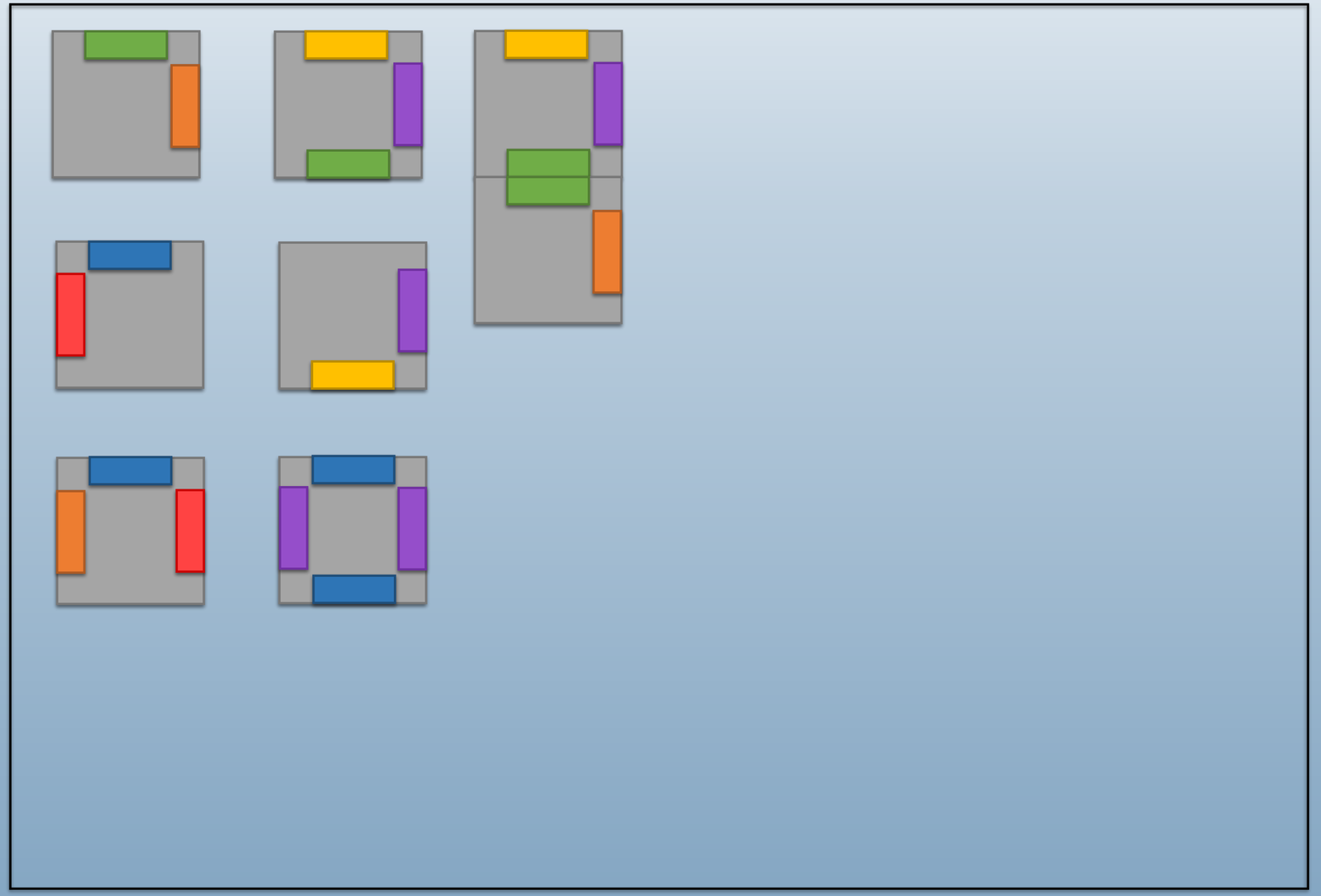
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

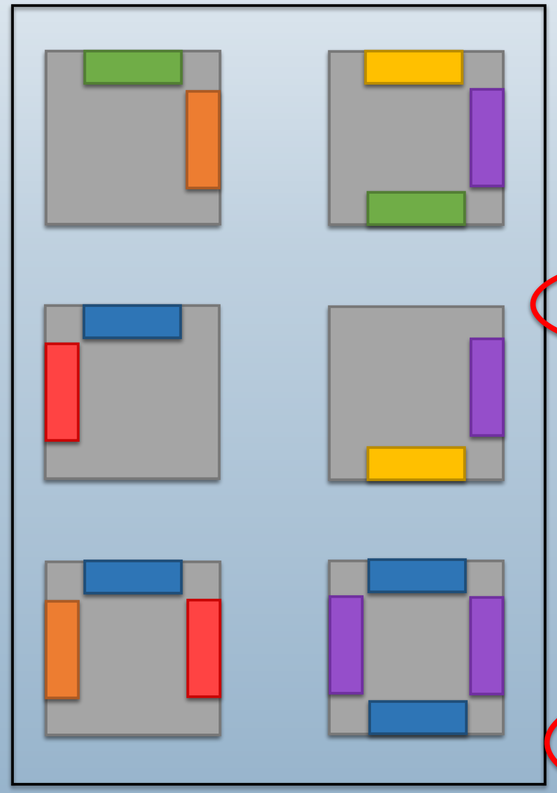
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

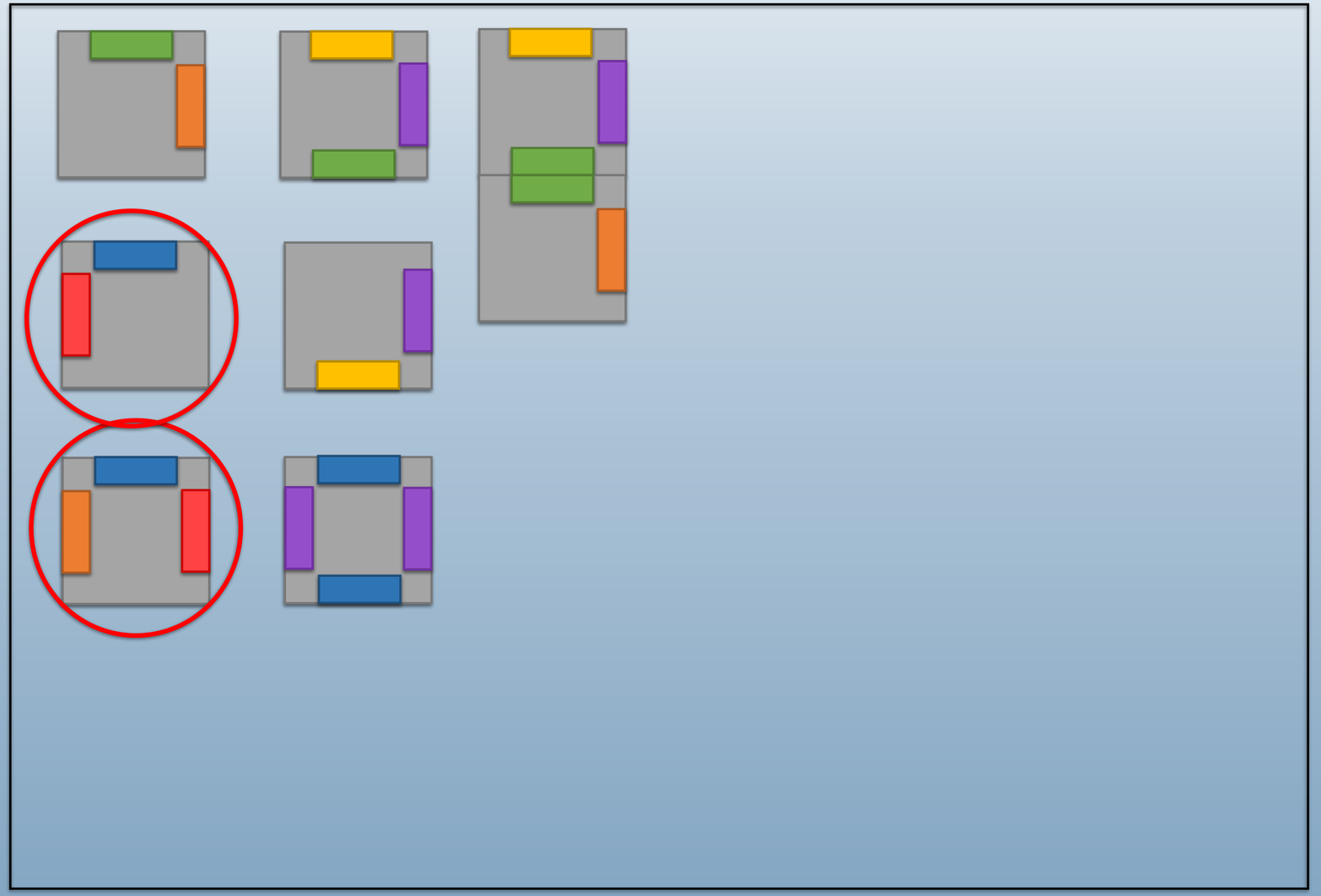
G(r) = 2

G(b) = 1

G(p) = 1

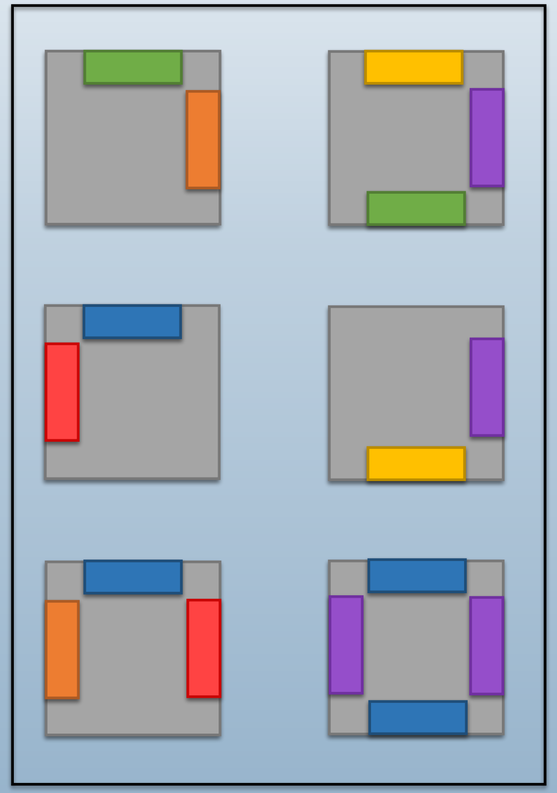
Temp. = 2

Producible Assemblies:



Tile Self-Assembly
(Rothemund, Winfree, Adleman)

Tileset:

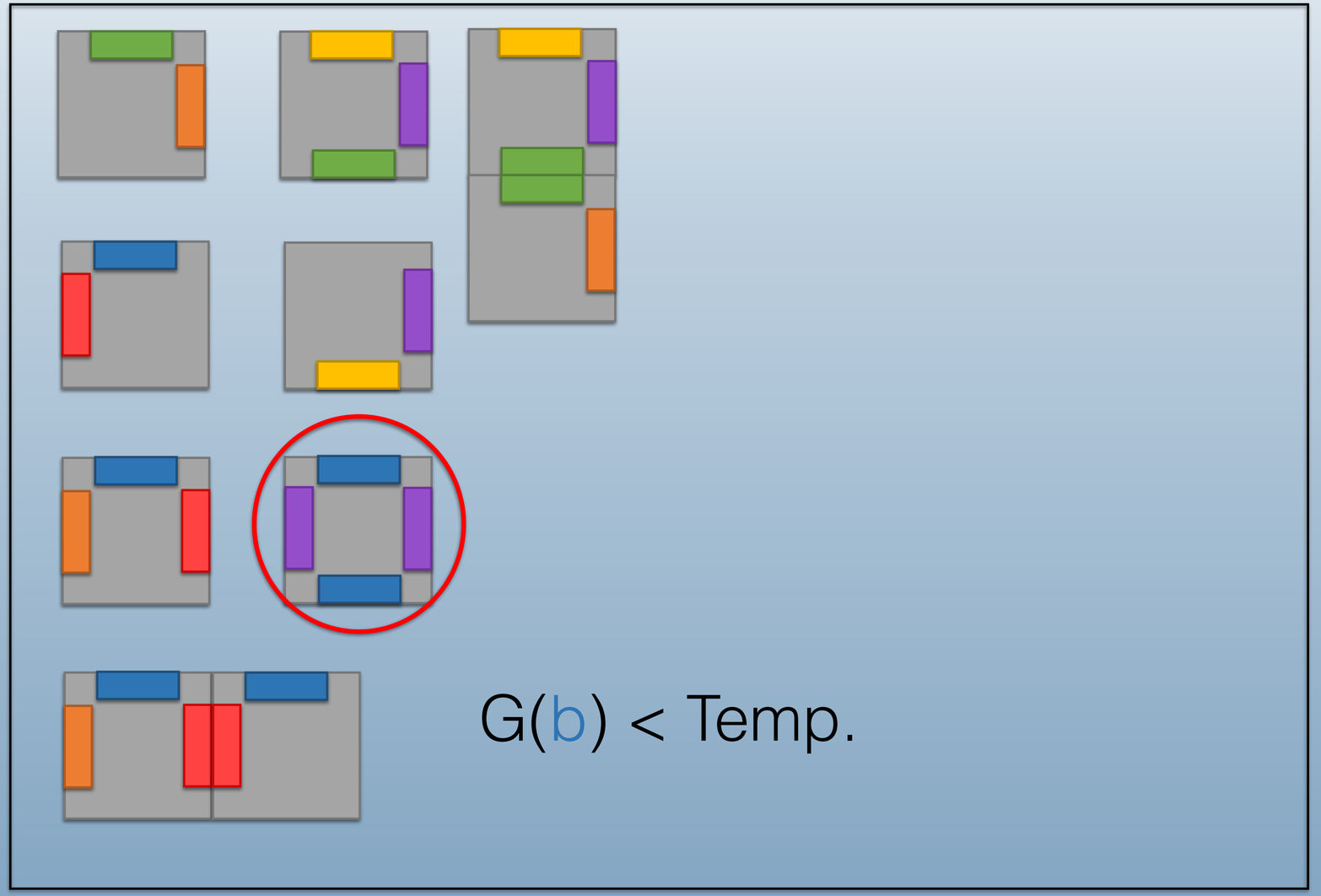


Glues:

G(g) = 2
G(o) = 2
G(y) = 2
G(r) = 2
G(b) = 1
G(p) = 1

Temp. = 2

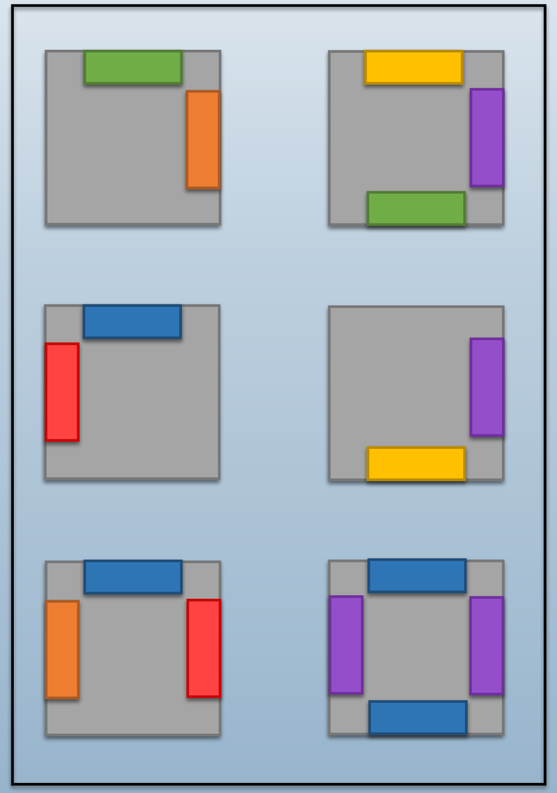
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

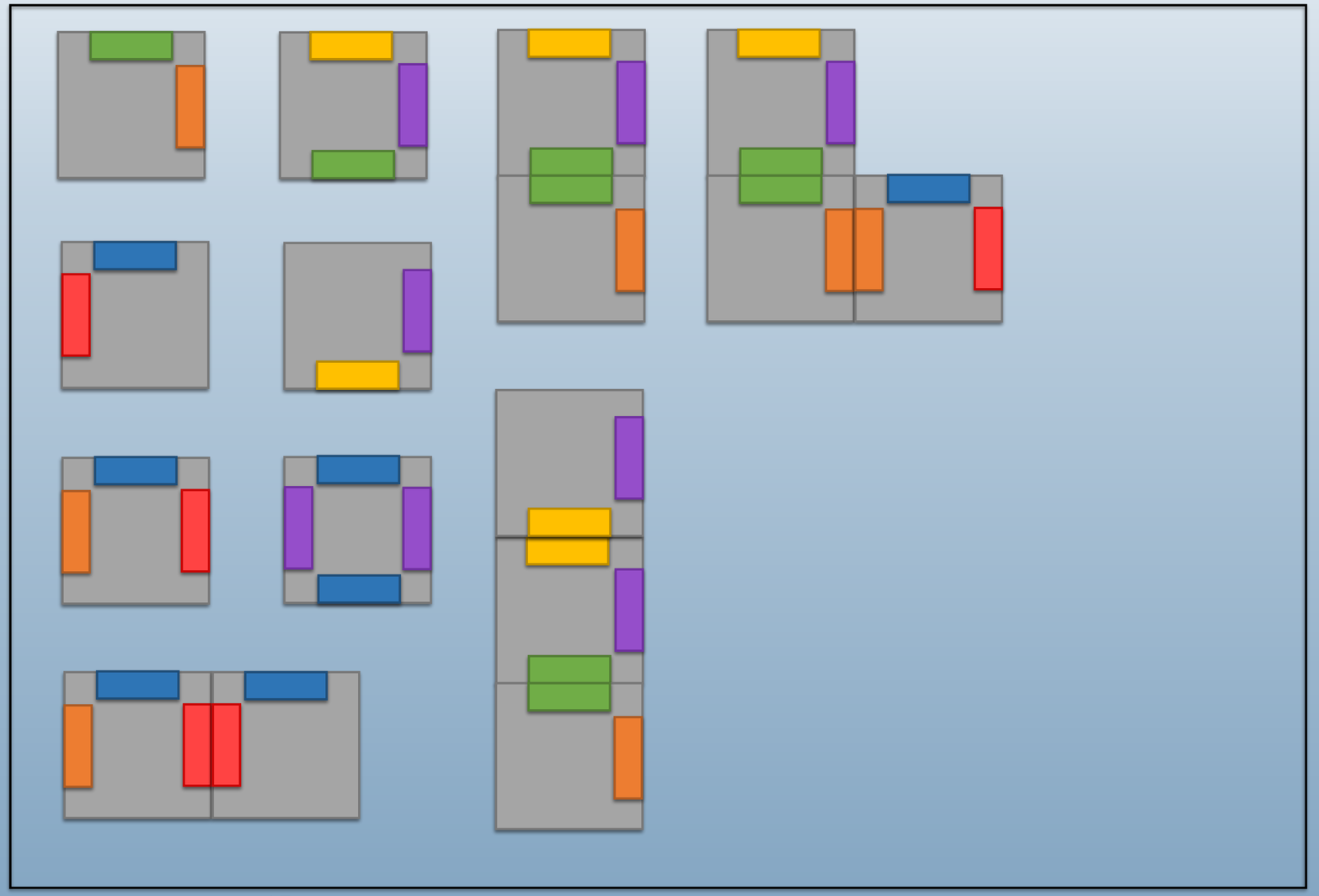
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

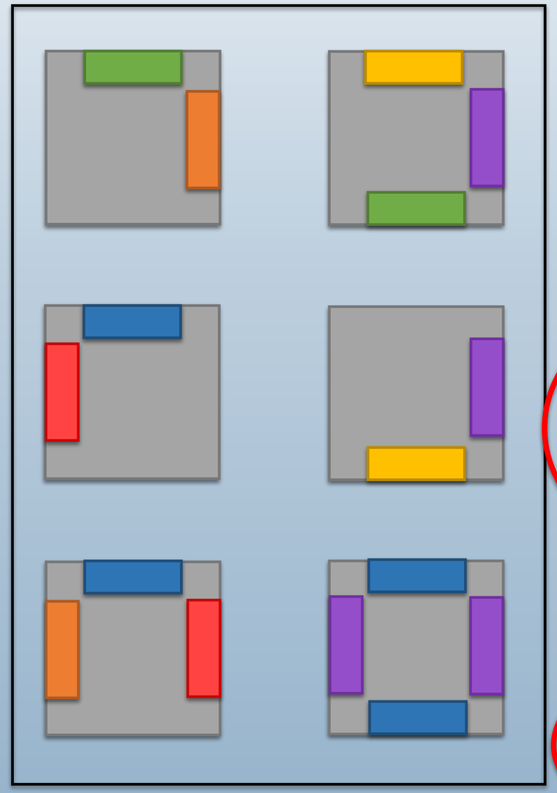
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

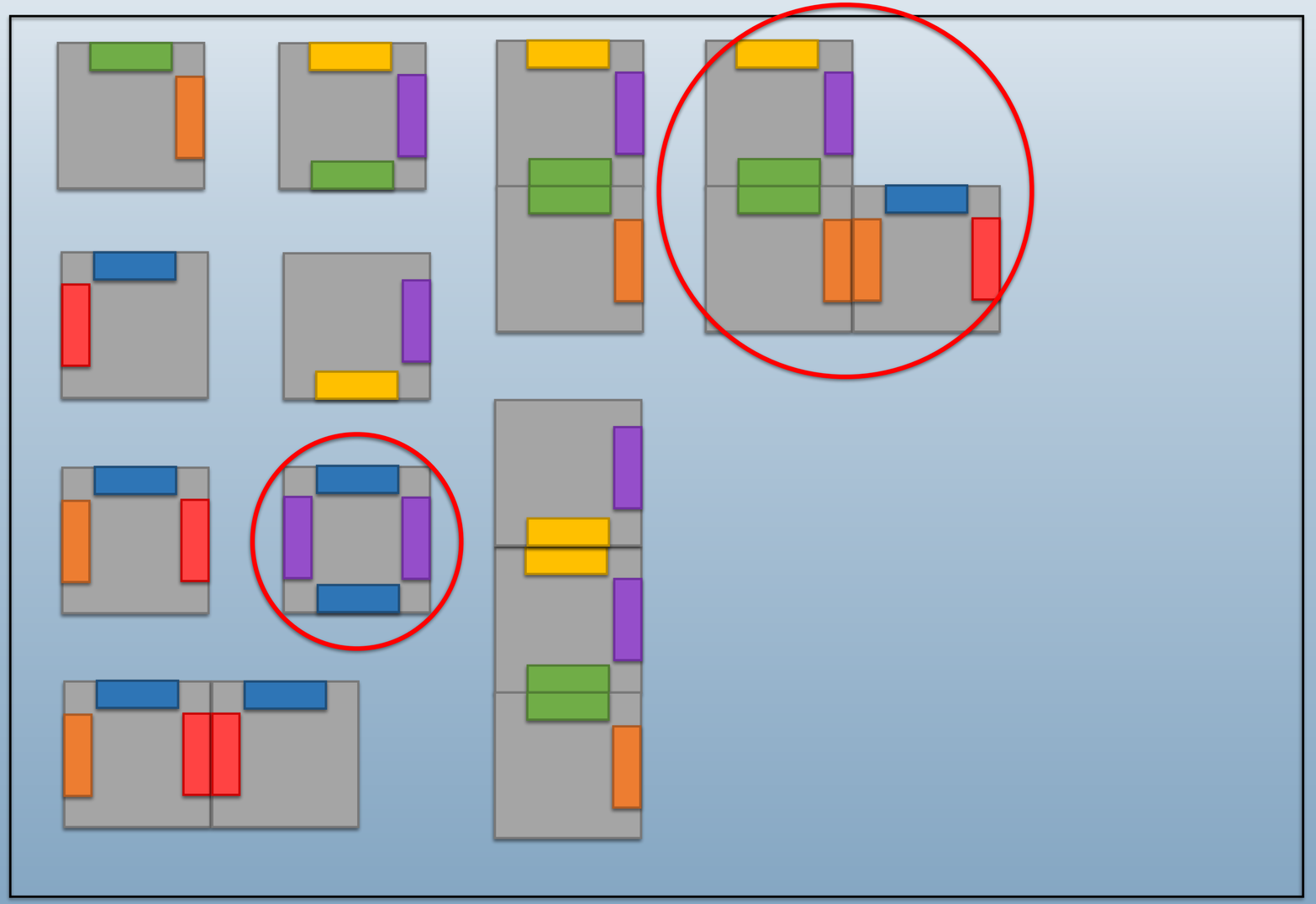
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

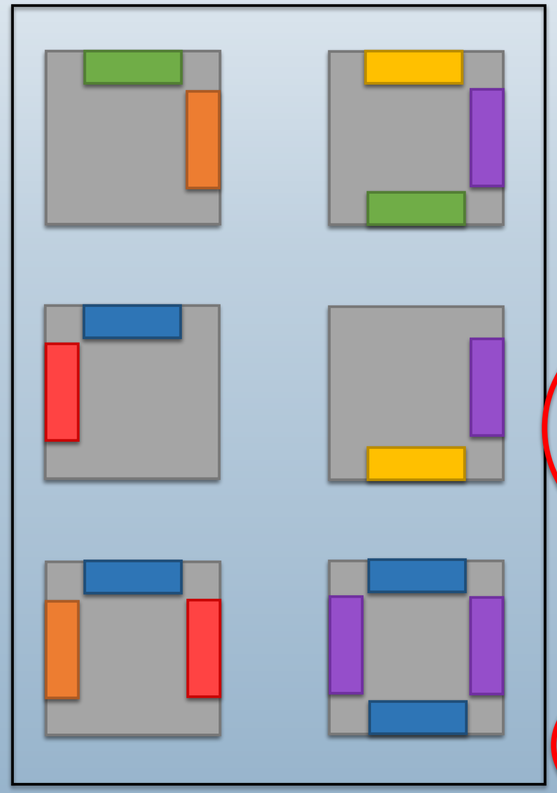
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

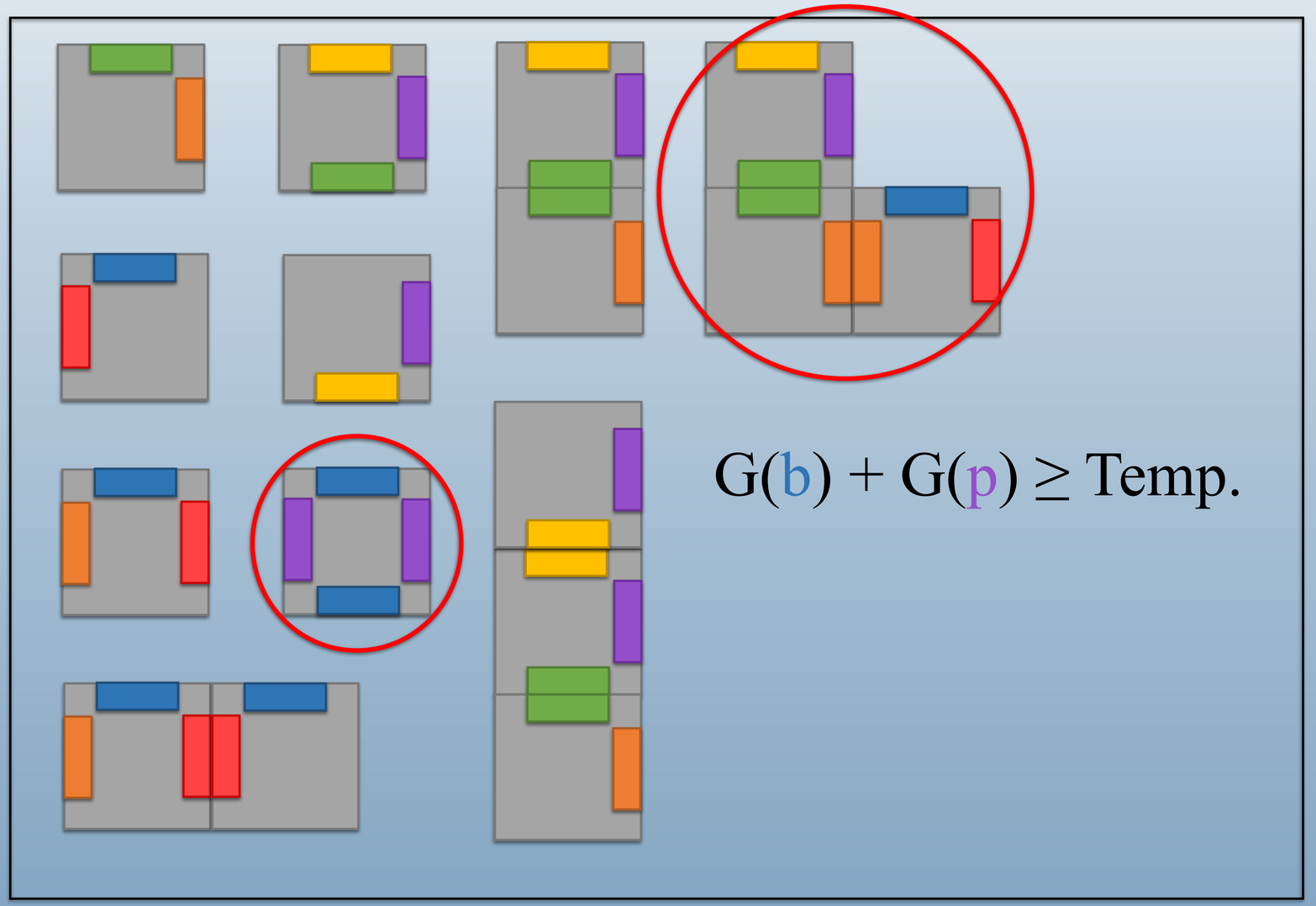
G(r) = 2

G(b) = 1

G(p) = 1

Temp. = 2

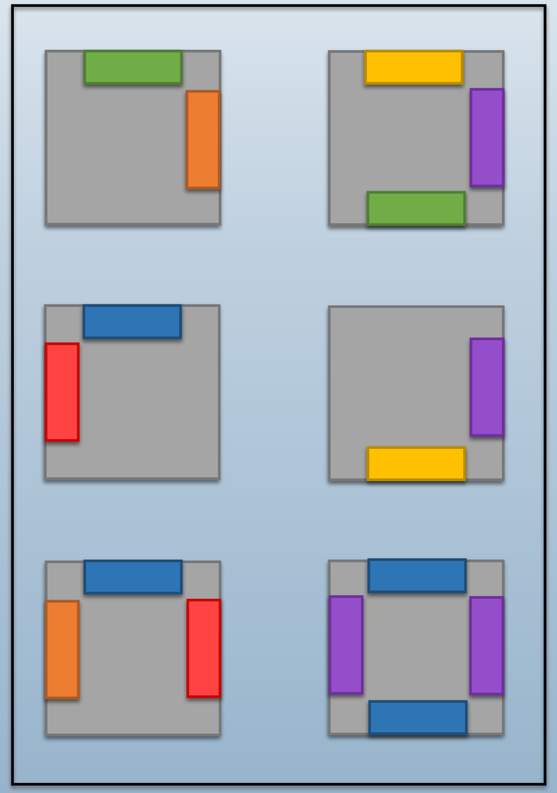
Producible Assemblies:



Tile Self-Assembly

(Rothemund, Winfree, Adleman)

Tileset:



Glues:

G(g) = 2

G(o) = 2

G(y) = 2

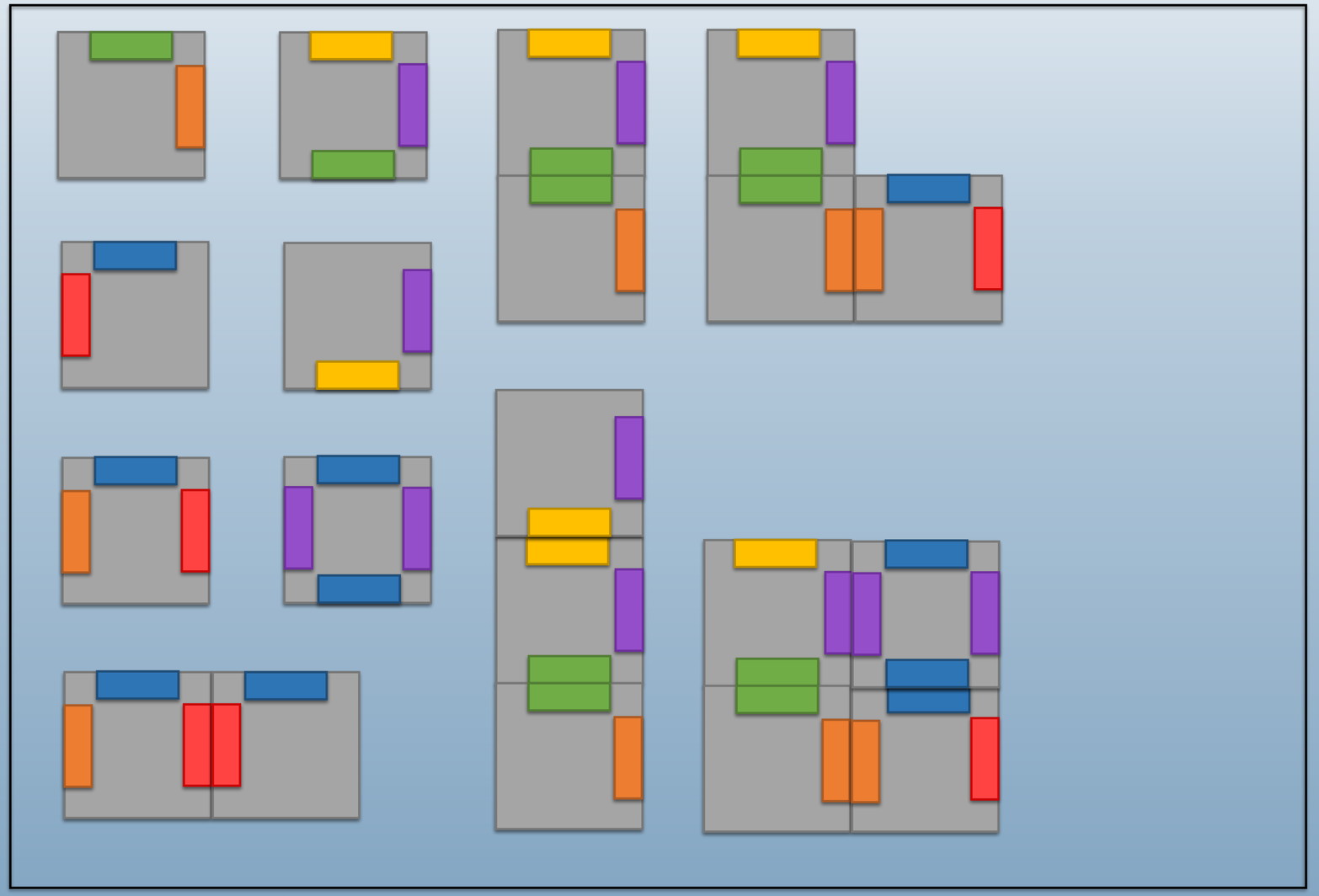
G(r) = 2

G(b) = 1

G(p) = 1

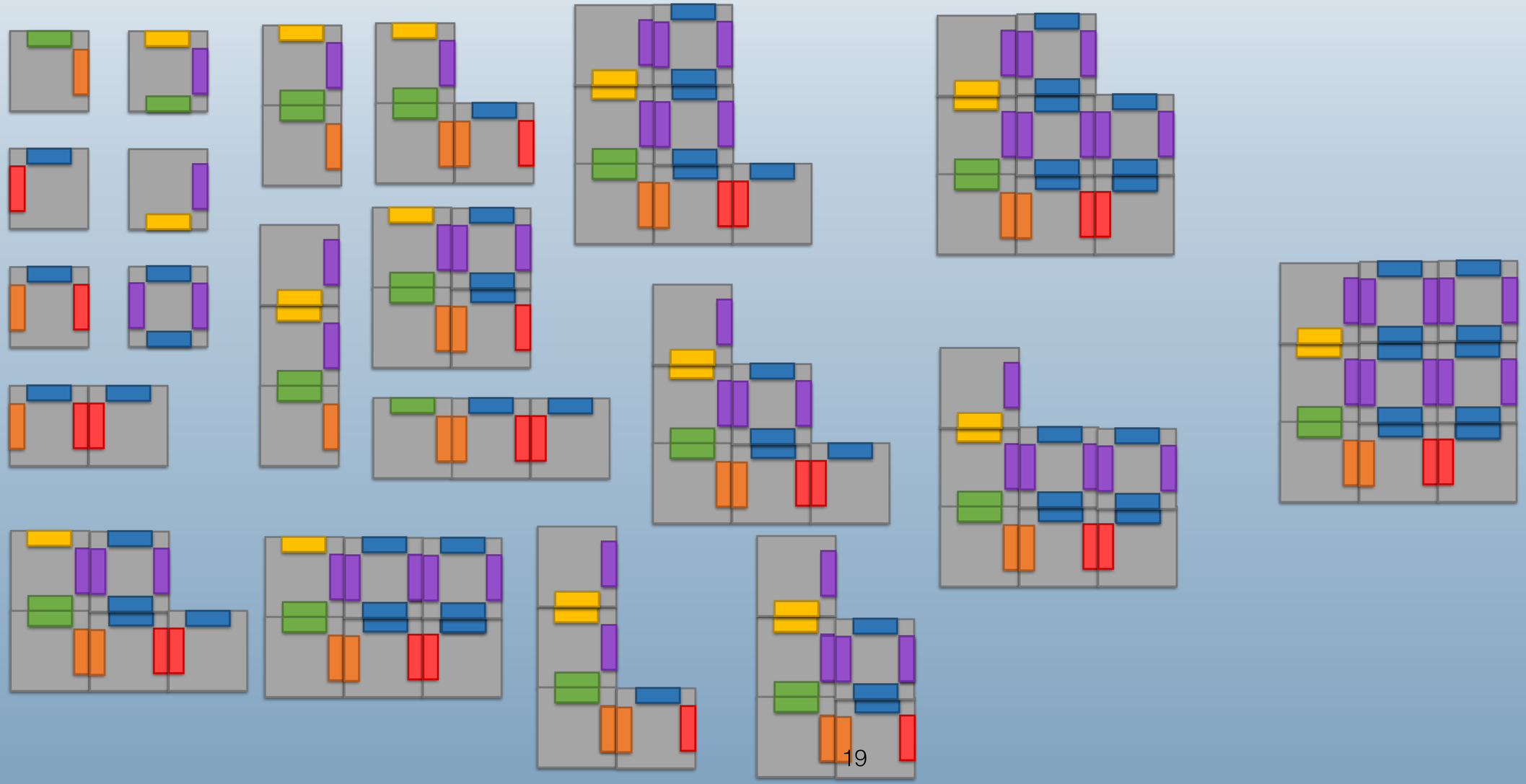
Temp. = 2

Producible Assemblies:



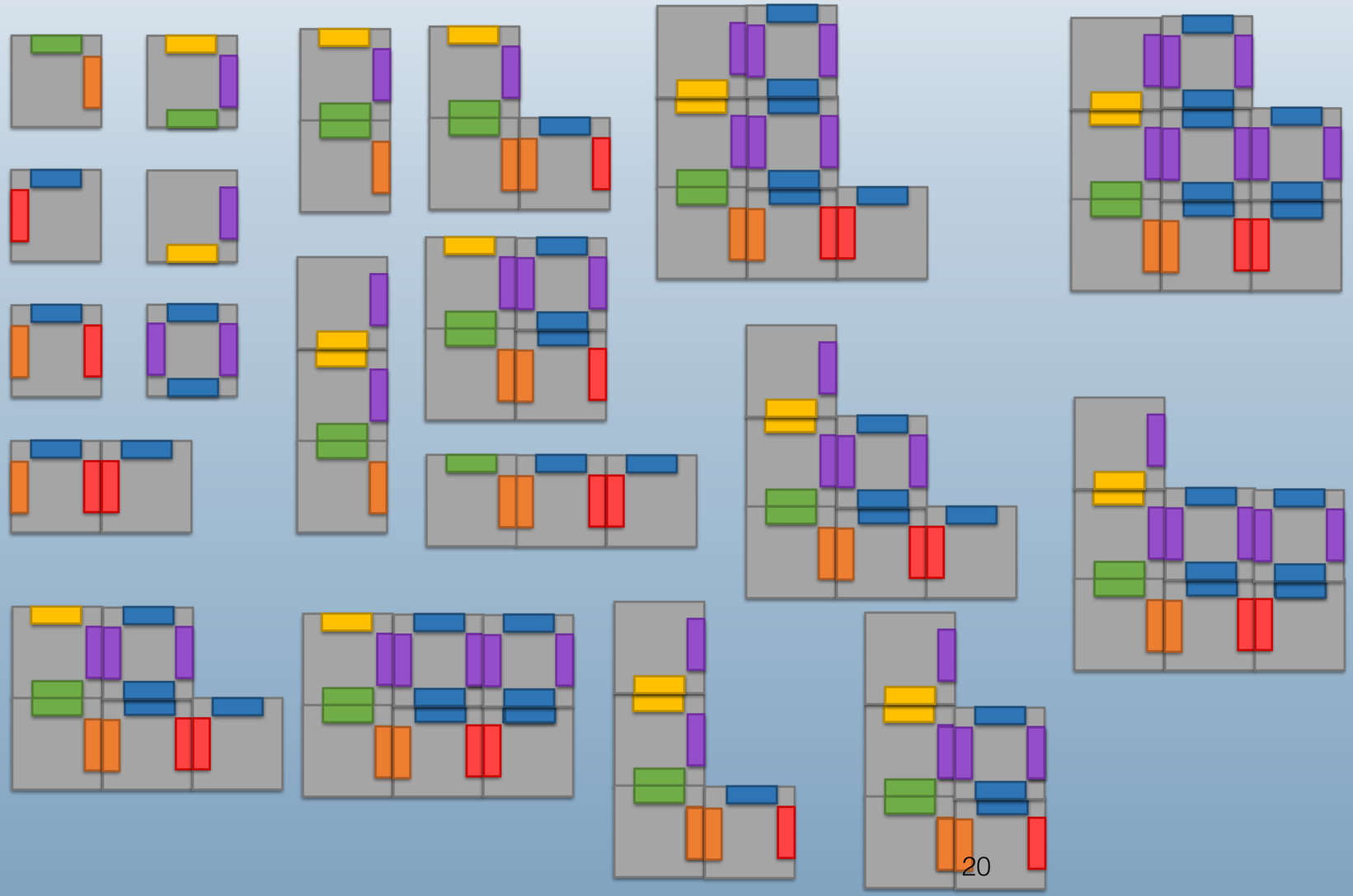
Tile Self-Assembly
(Rothemund, Winfree, Adleman)

Producible Assemblies:

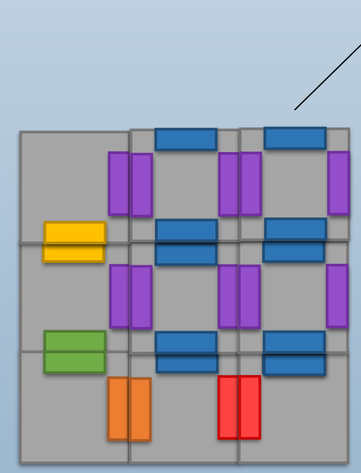


Tile Self-Assembly
(Rothemund, Winfree, Adleman)

Producible Assemblies:

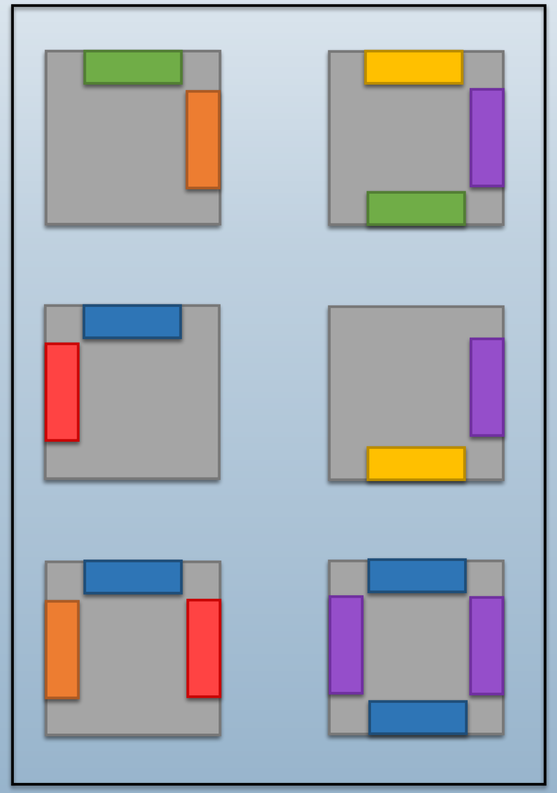


Terminal



Tile Self-Assembly
(Rothemund, Winfree, Adleman)

Tileset:

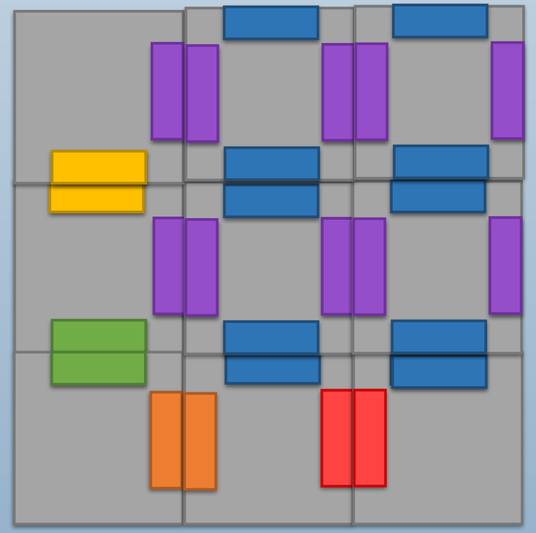


Glues:

G(g) = 2
G(o) = 2
G(y) = 2
G(r) = 2
G(b) = 1
G(p) = 1

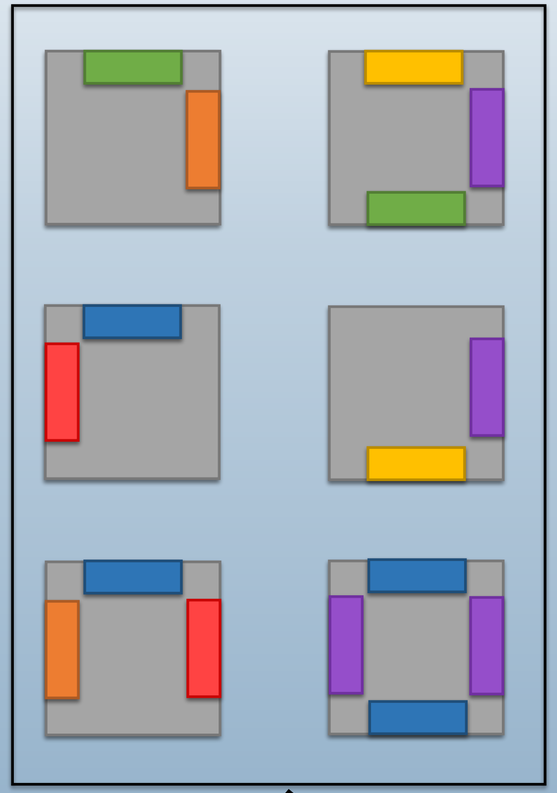
Temp. = 2

This system produces a 3x3 square:



Tile Self-Assembly
(Rothemund, Winfree, Adleman)

Tileset:



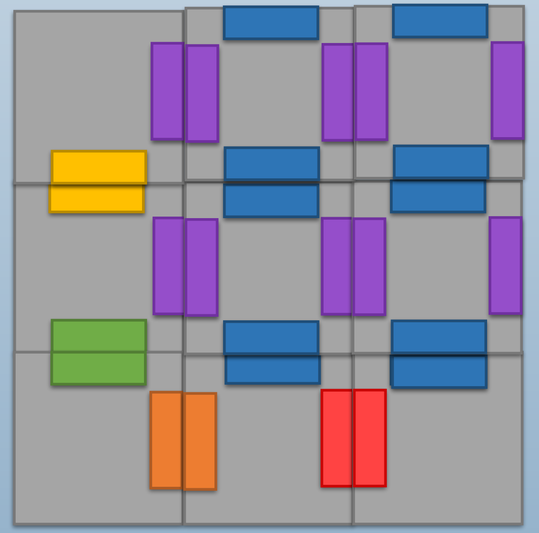
Glues:

G(g) = 2
G(o) = 2
G(y) = 2
G(r) = 2
G(b) = 1
G(p) = 1

Temp. = 2

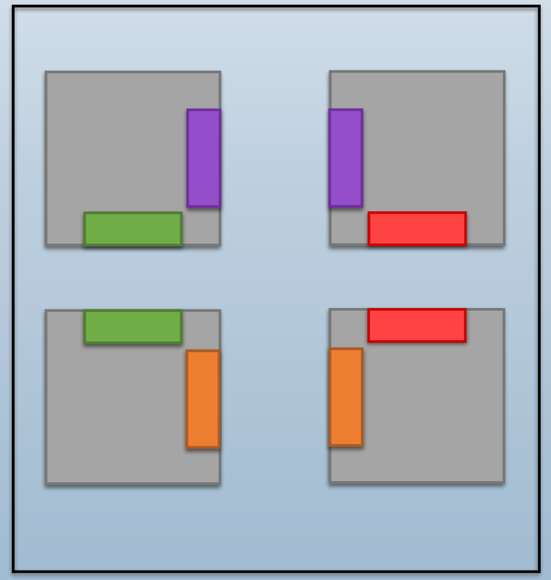
Tile Complexity

This system produces a 3x3 square:



The 2HAM with Negative Glues

Tileset:



Glues:

G(g) = 2

G(p) = 2

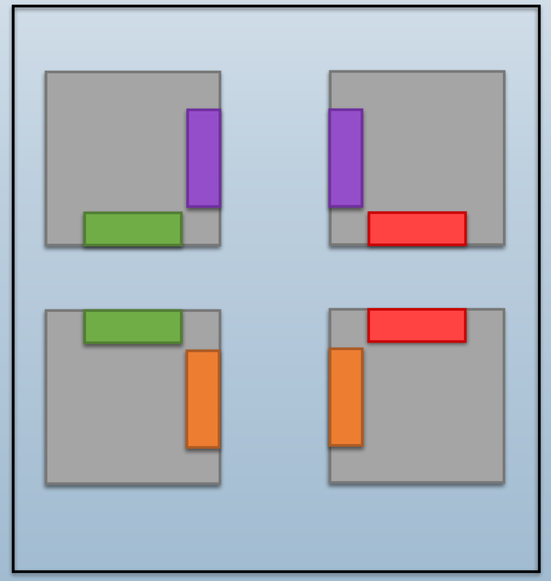
G(o) = 1

G(r)= -1

Temp. = 2

The 2HAM with Negative Glues

Tileset:



Glues:

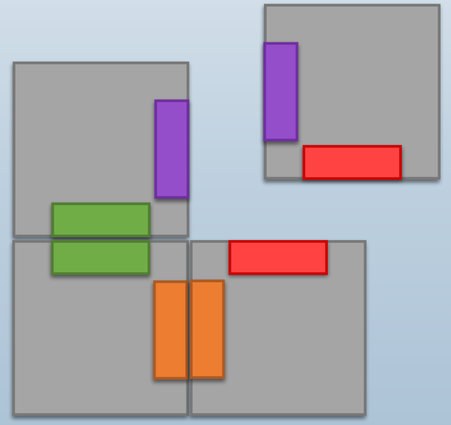
G(g) = 2

G(p) = 2

G(o) = 1

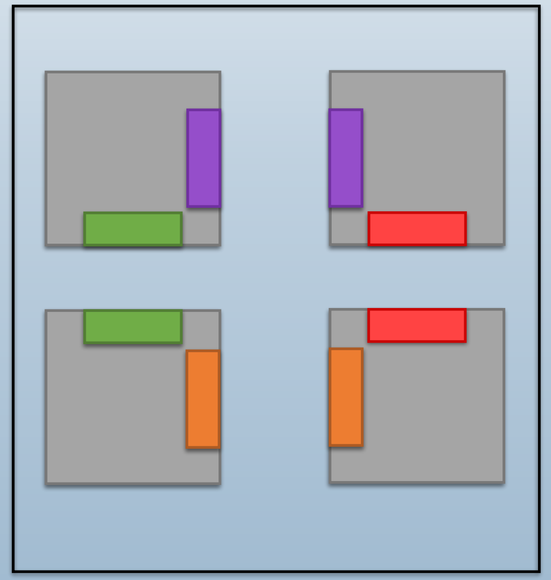
G(r)= -1

Temp. = 1



The 2HAM with Negative Glues

Tileset:



Glues:

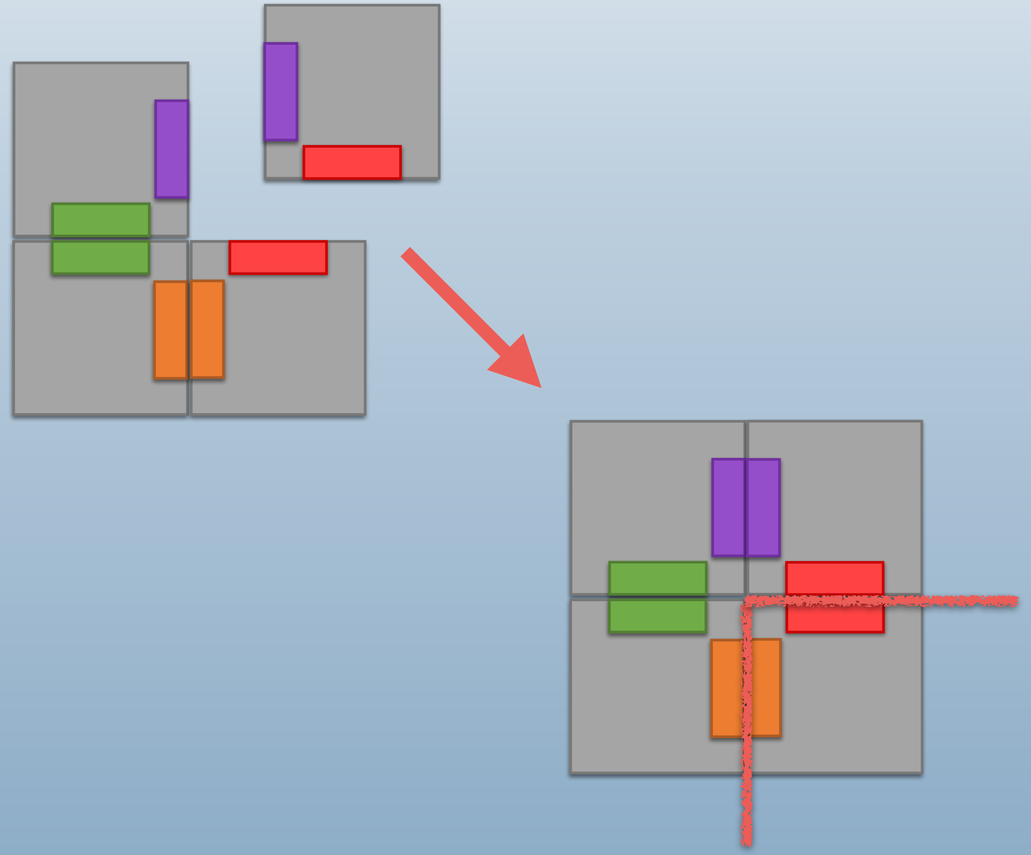
G(g) = 2

G(p) = 2

G(o) = 1

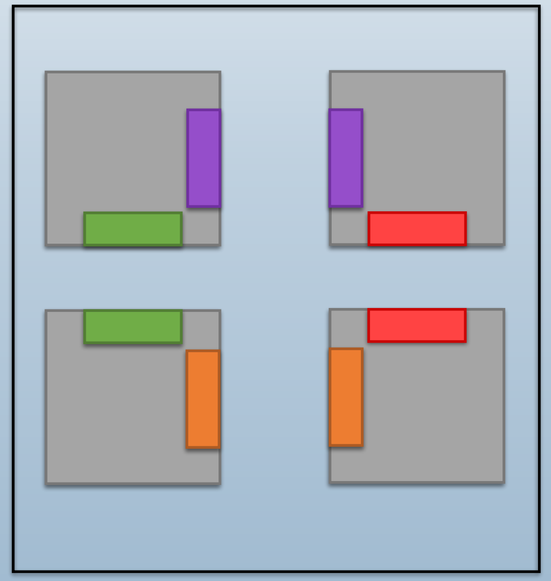
G(r)= -1

Temp. = 1



The 2HAM with Negative Glues

Tileset:



Glues:

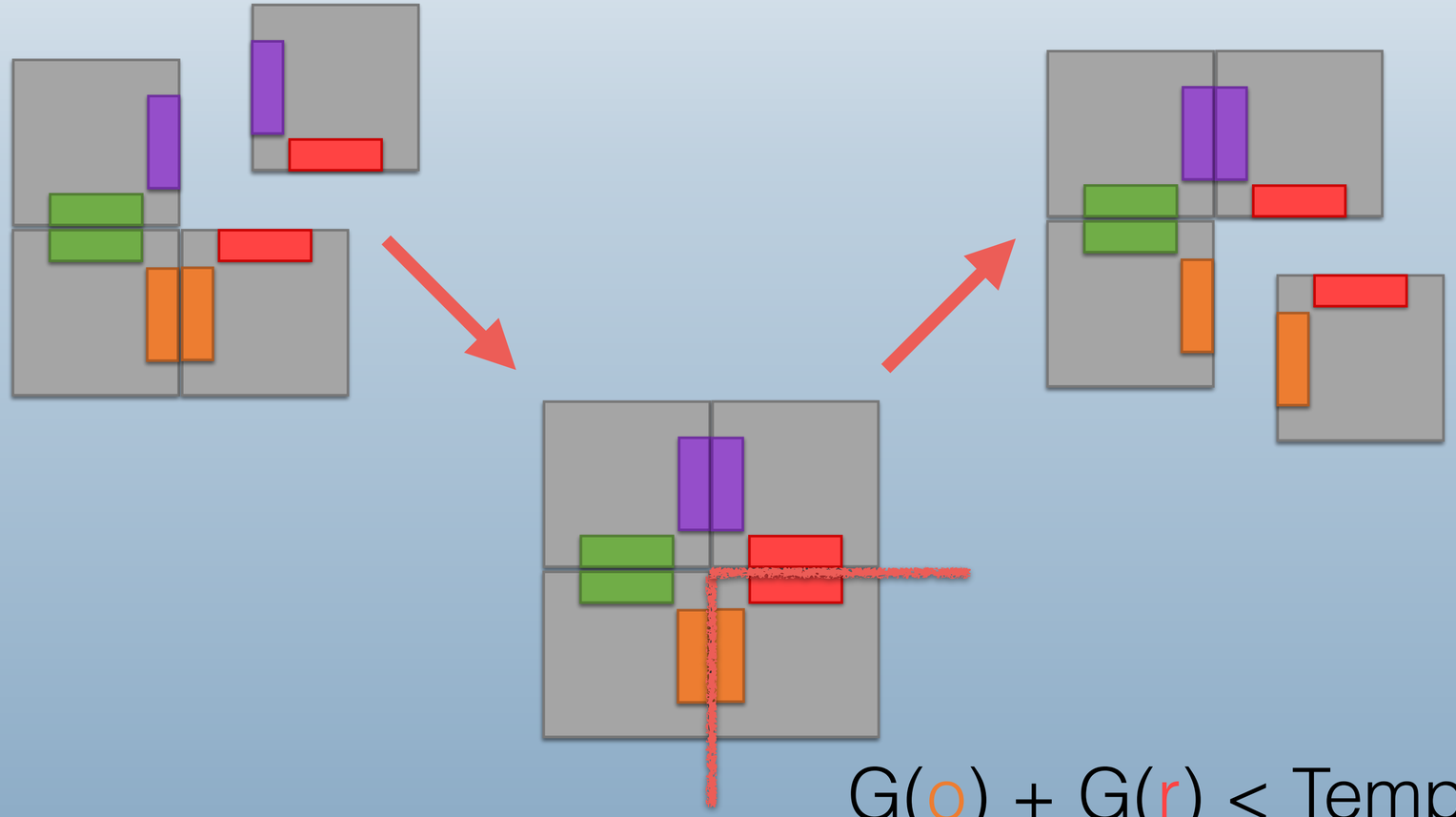
G(g) = 2

G(p) = 2

G(o) = 1

G(r)= -1

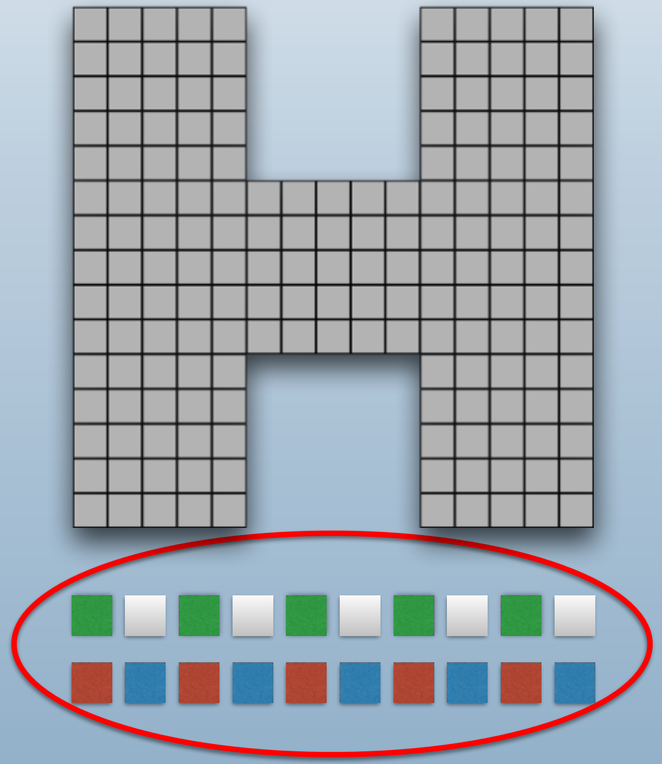
Temp. = 1



G(o) + G(r) < Temp.

Replication

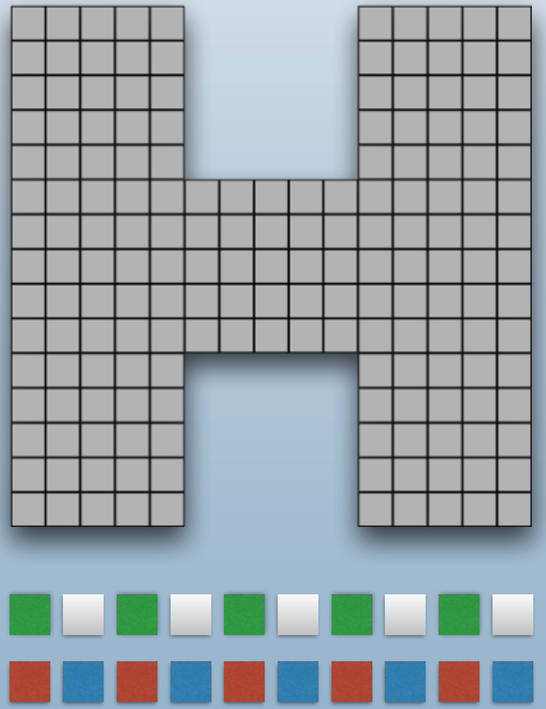
Input



Input tileset

Replication

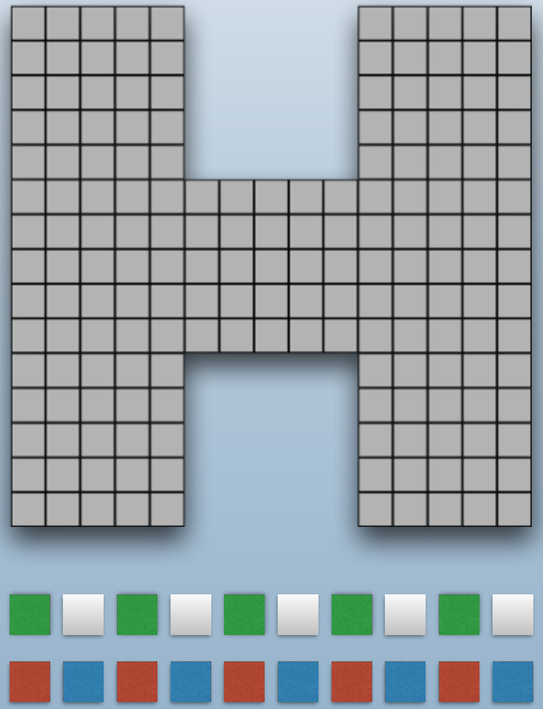
Input



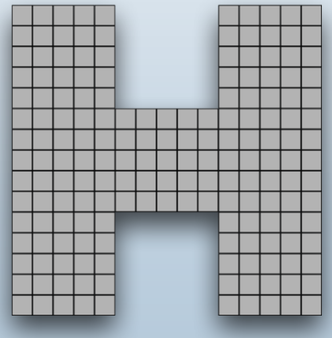
The shape to replicate

Replication

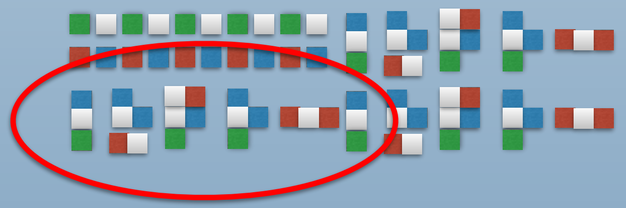
Input



Output



Producible assemblies of the tileset alone

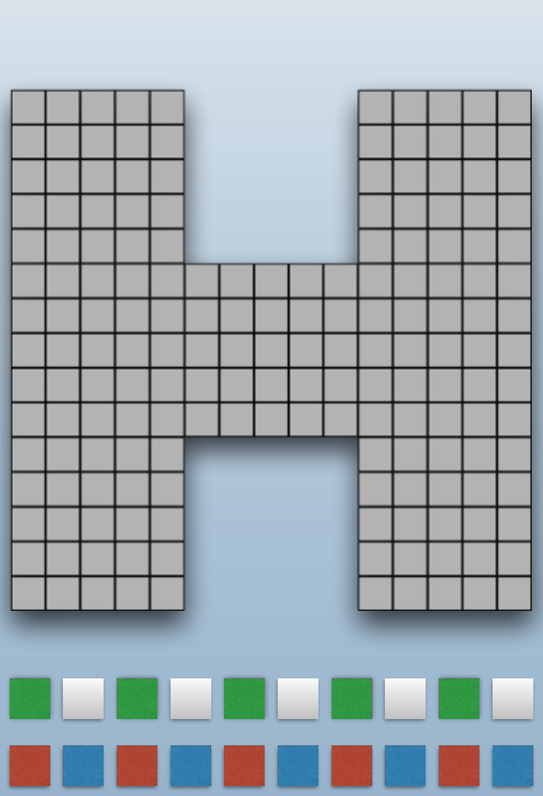


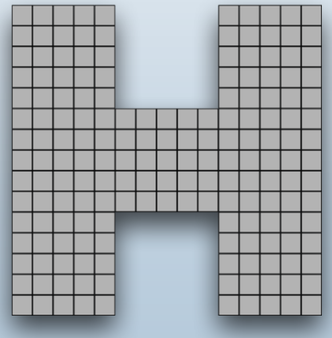
...

Replication

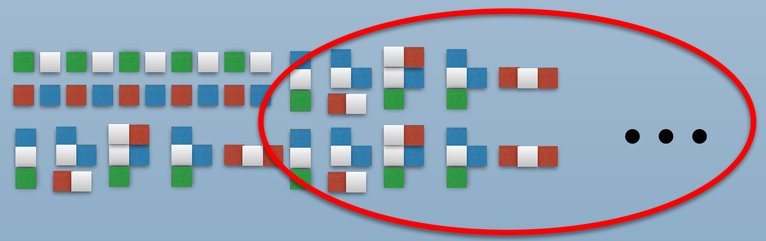
Input

Output





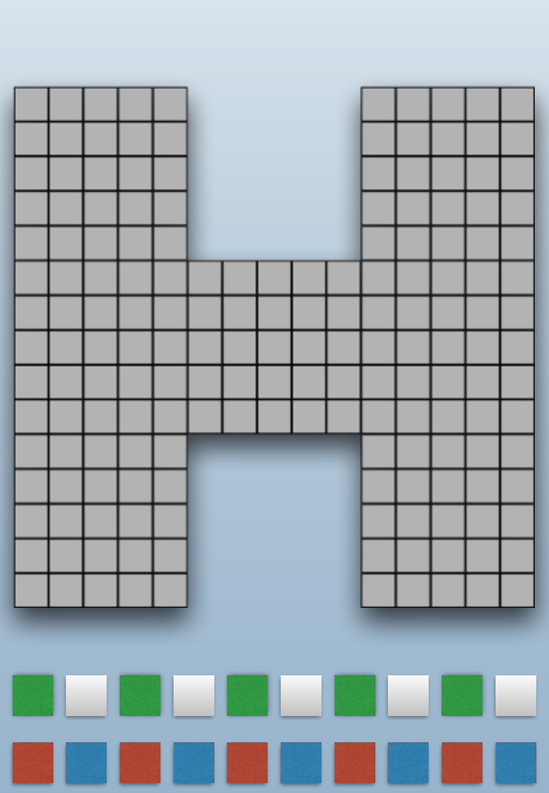
Additional producible assemblies that can only be created if the shape is in the tile system



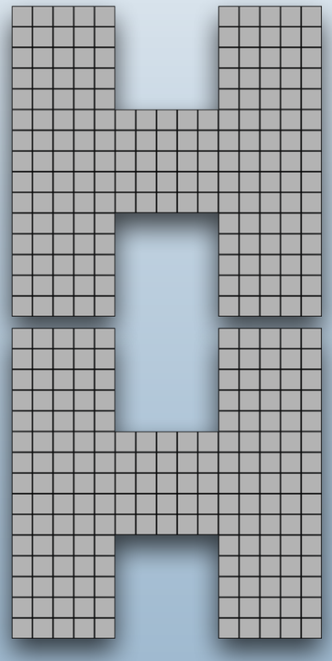
Replication

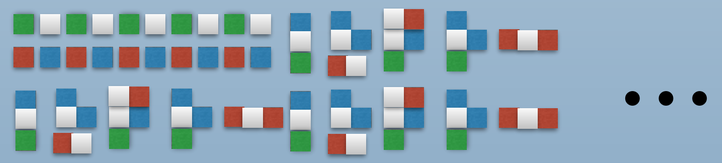
Input

Output





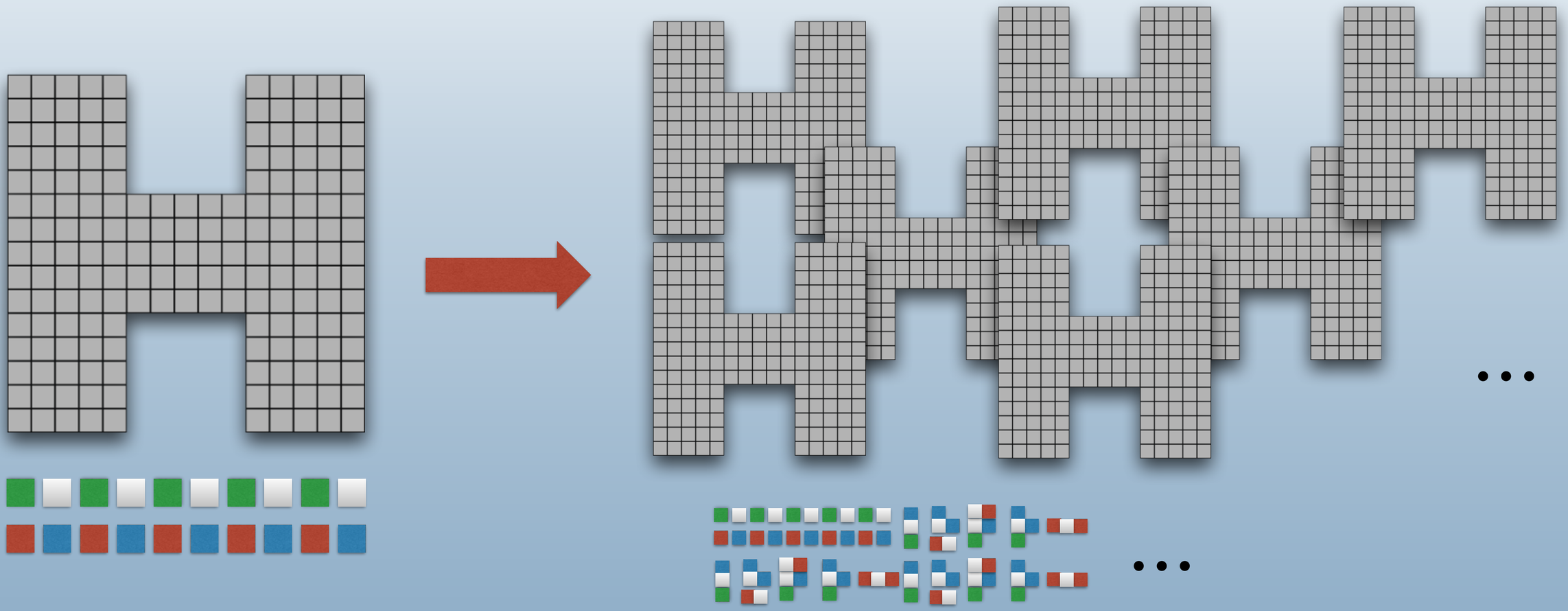




Replication

Input

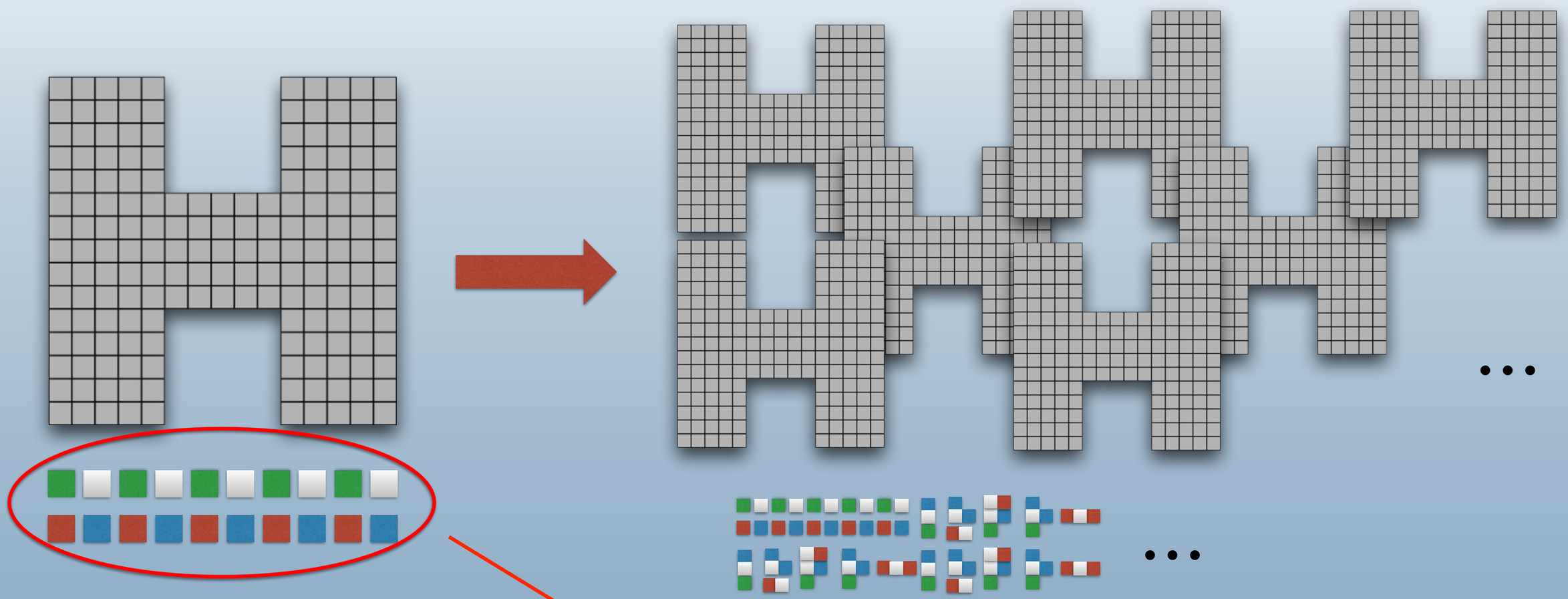
Output



Replication

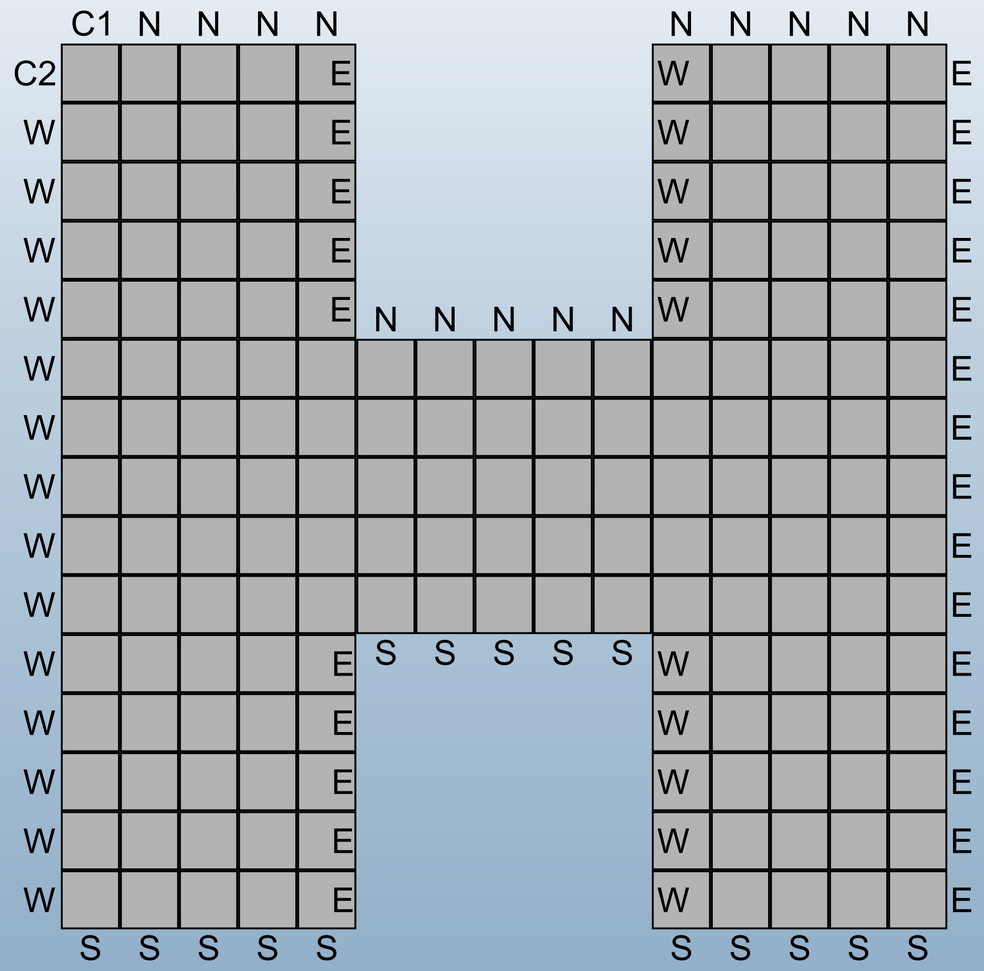
Input

Output



Input tileset = replication system

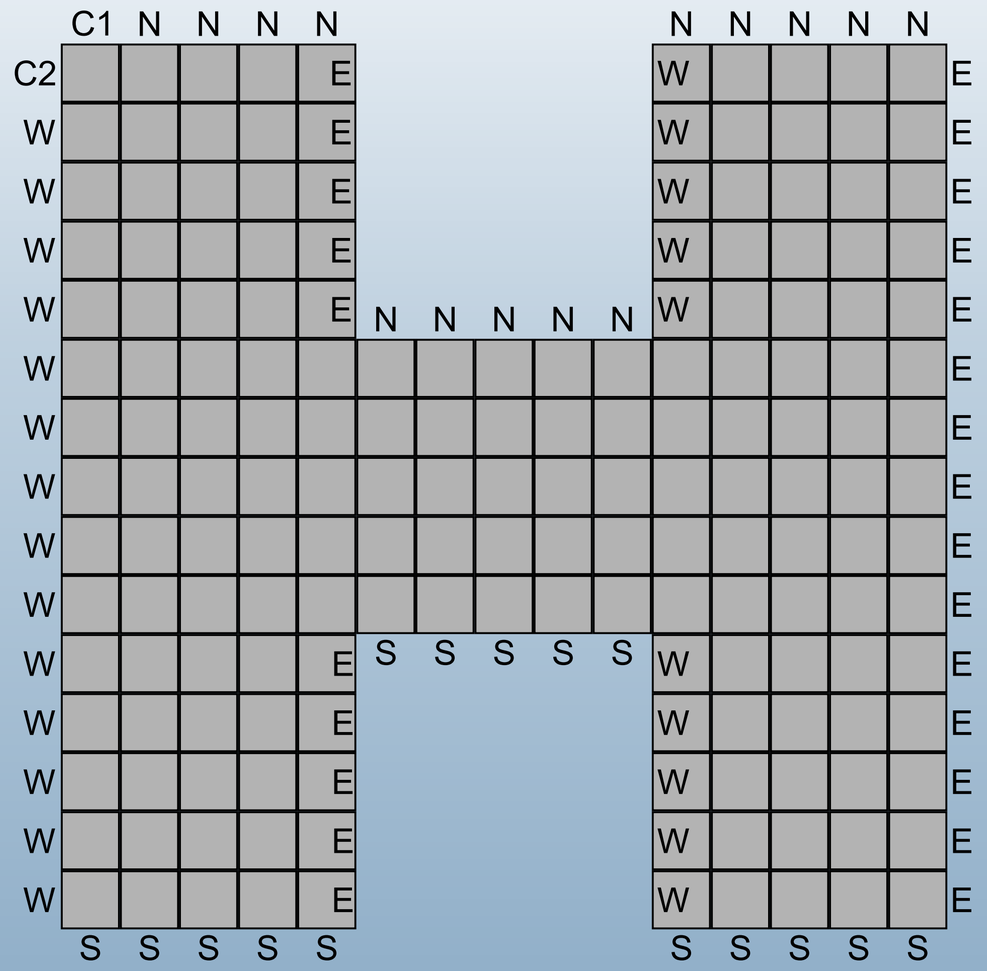
Definitions / Results



Universal Shape Replicator

- Any super-constant size assembly is the target shape (or will be eventually)
- Unbounded replication of shape

Definitions / Results



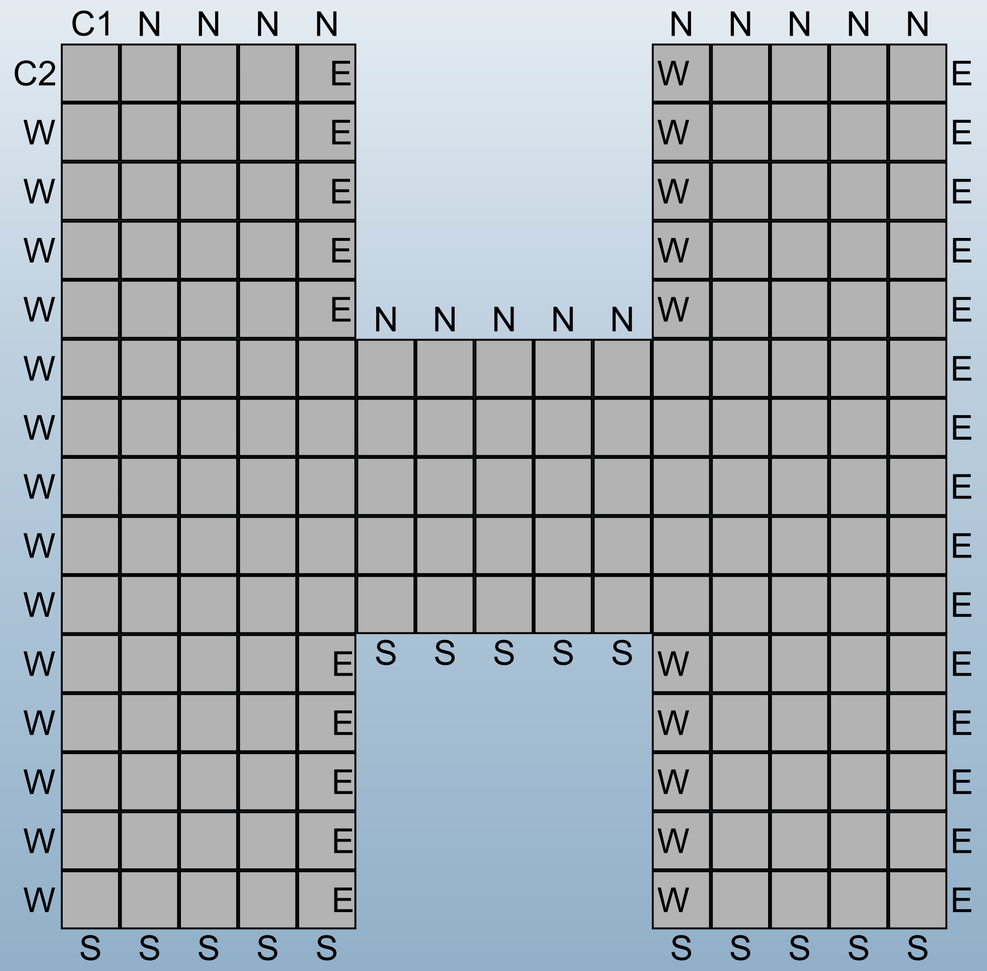
Universal Shape Replicator

- Any super-constant size assembly is the target shape (or will be eventually)
- Unbounded replication of shape

A *Sleek* replicator

- Constant input tile-set size (replicator)
- All producible assemblies of size $O(X)$ for shape X
- Only produces constant-size garbage
- Simple shape constraints
  - Constant Minimum feature size
  - Uniform glues with $O(1)$ exception

Definitions / Results



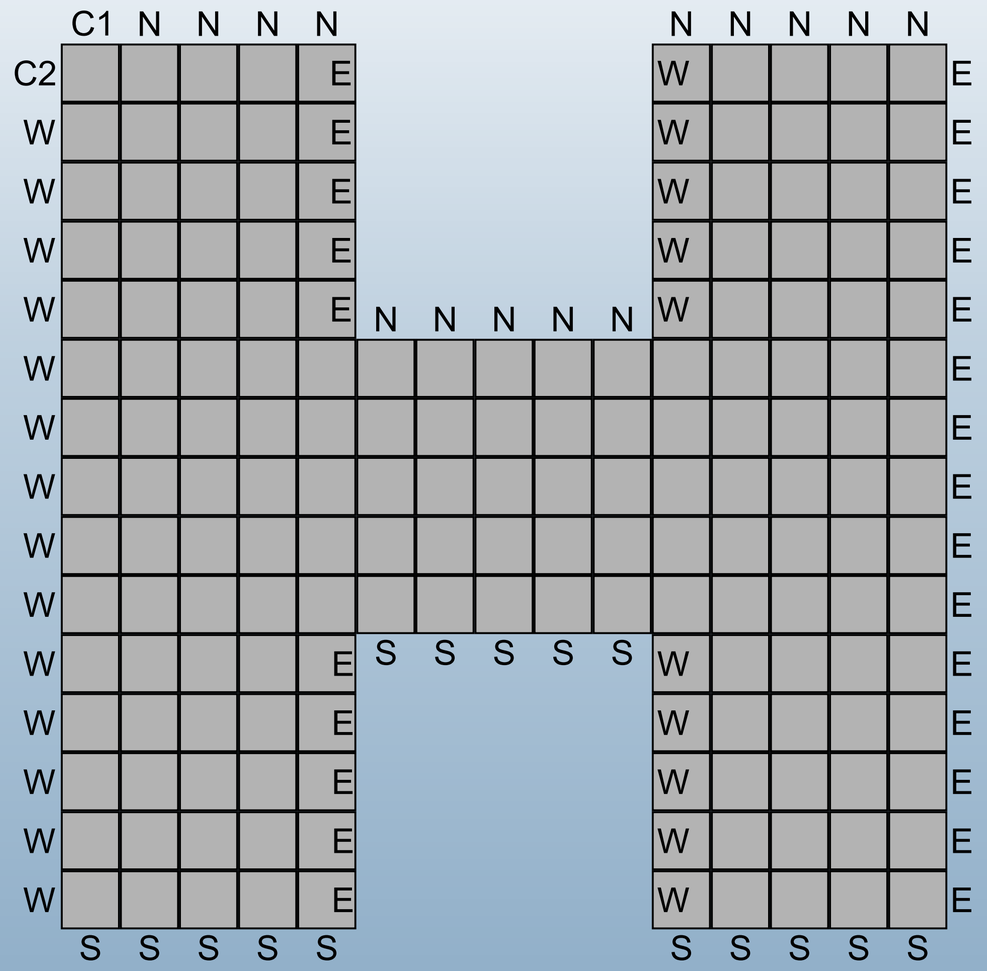
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Definitions / Results



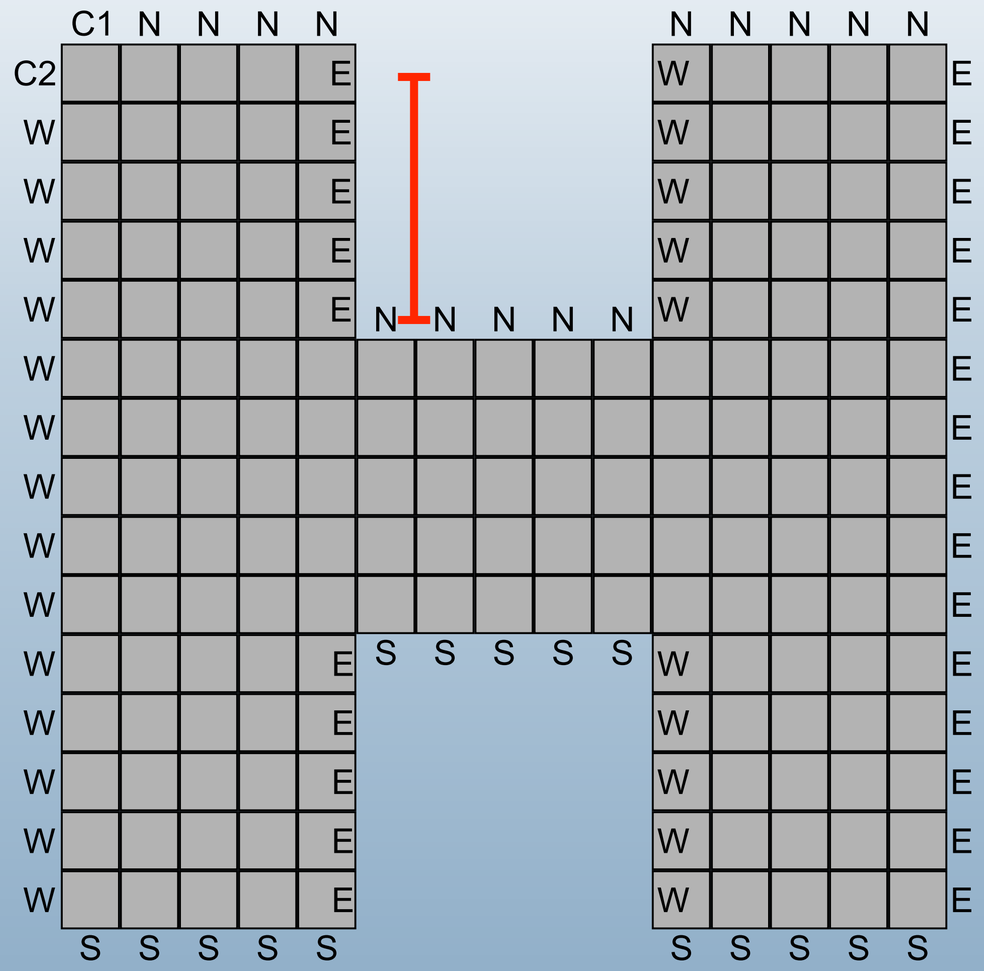
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Definitions / Results



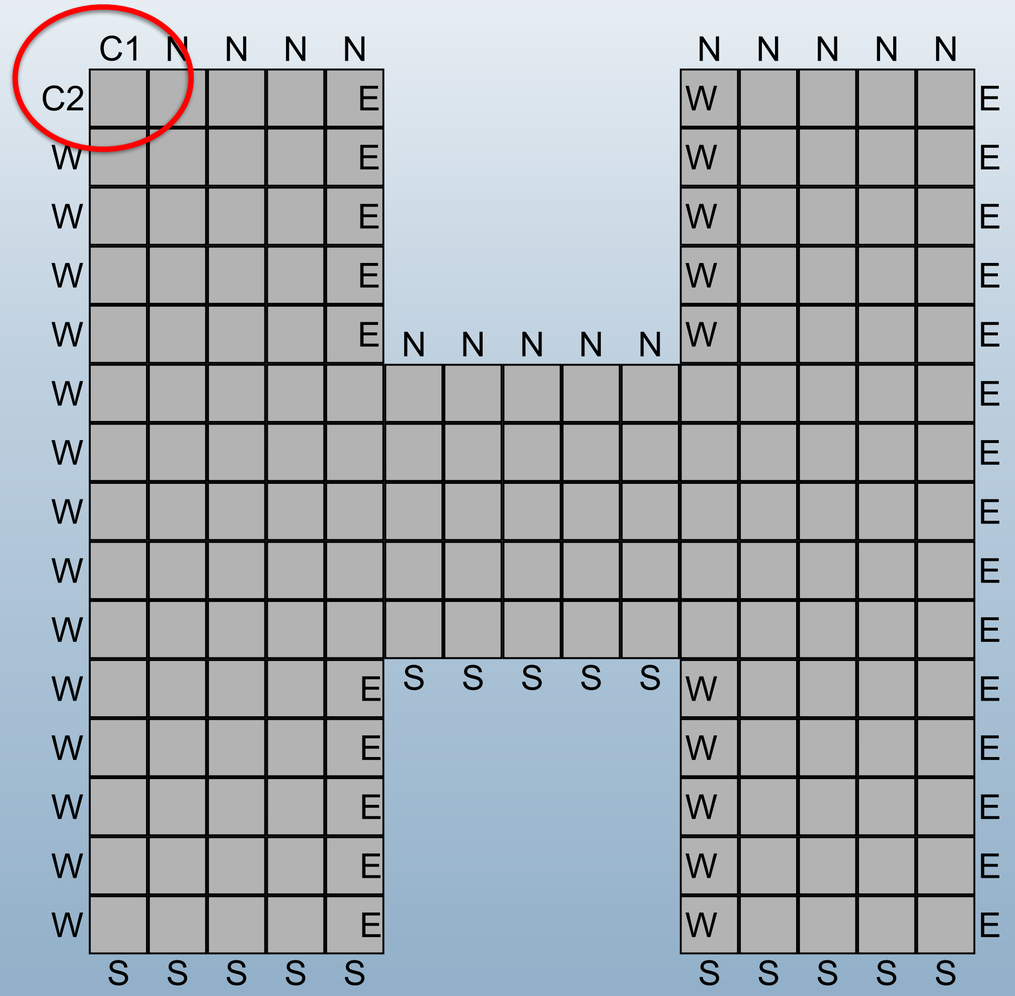
Universal Shape Replicator

- Any super-constant size assembly is the target shape (or will be eventually)
- Unbounded replication of shape

A *Sleek* replicator

- Constant input tile-set size (replicator)
- All producible assemblies of size $O(X)$ for shape X
- Only produces constant-size garbage
- Simple shape constraints
  - Constant Minimum feature size 9
  - Uniform glues with $O(1)$ exception

Definitions / Results



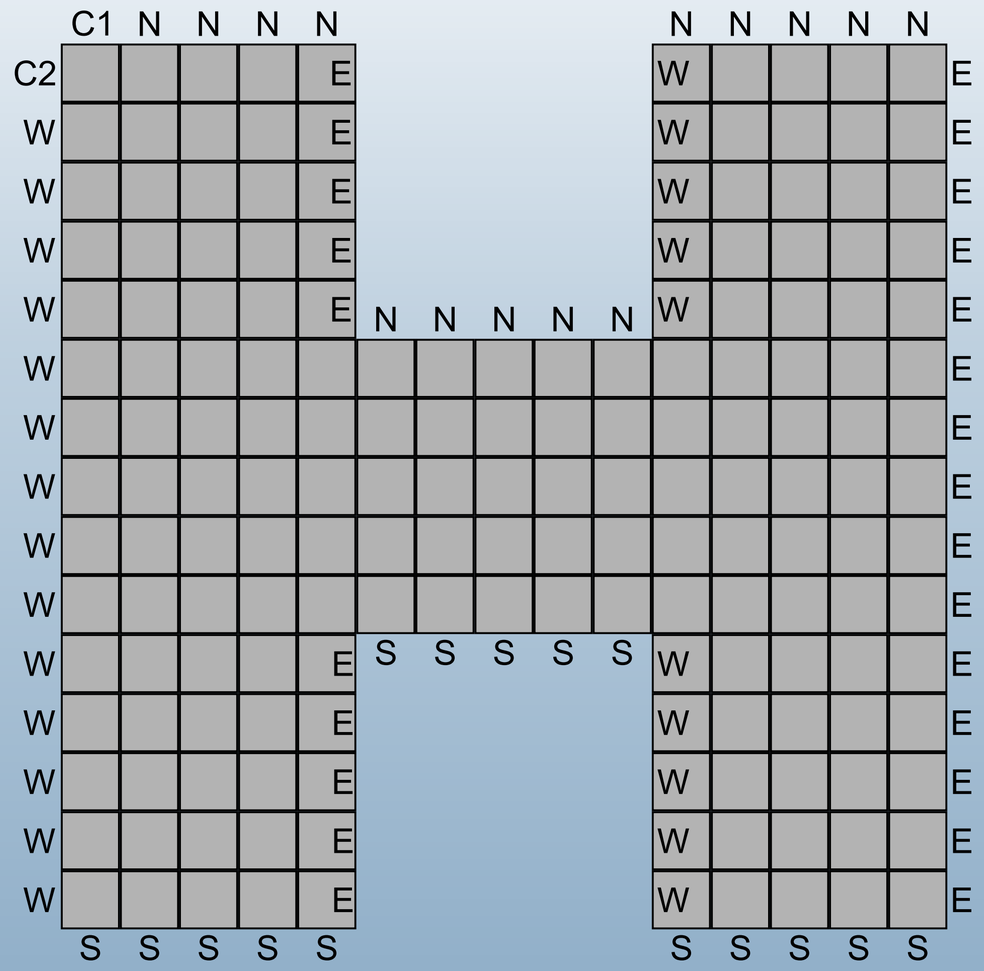
Universal Shape Replicator

- Any super-constant size assembly is the target shape (or will be eventually)
- Unbounded replication of shape

A *Sleek* replicator

- Constant input tile-set size (replicator)
- All producible assemblies of size $O(X)$ for shape X
- Only produces constant-size garbage
- Simple shape constraints
  - Constant Minimum feature size — 9
  - Uniform glues with $O(1)$ exception — 1

Definitions / Results



Universal Shape Replicator

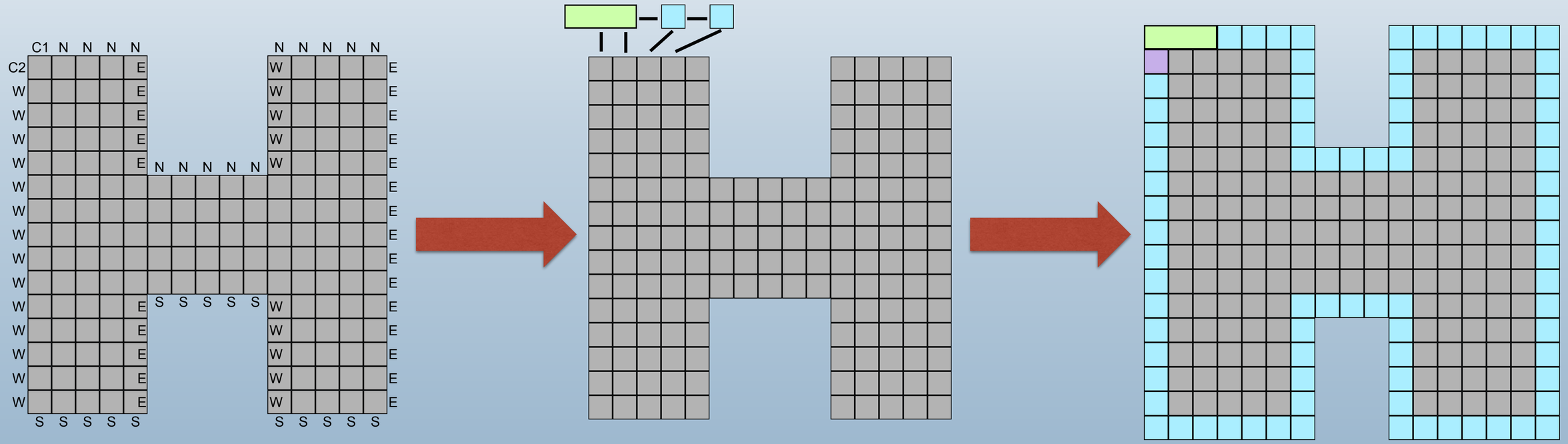
- Any super-constant size assembly is the target shape (or will be eventually)
- Unbounded replication of shape

A *Sleek* replicator

- Constant input tile-set size (replicator)
- All producible assemblies of size $O(X)$ for shape X
- Only produces constant-size garbage
- Simple shape constraints
  - Constant Minimum feature size
  - Uniform glues with $O(1)$ exception

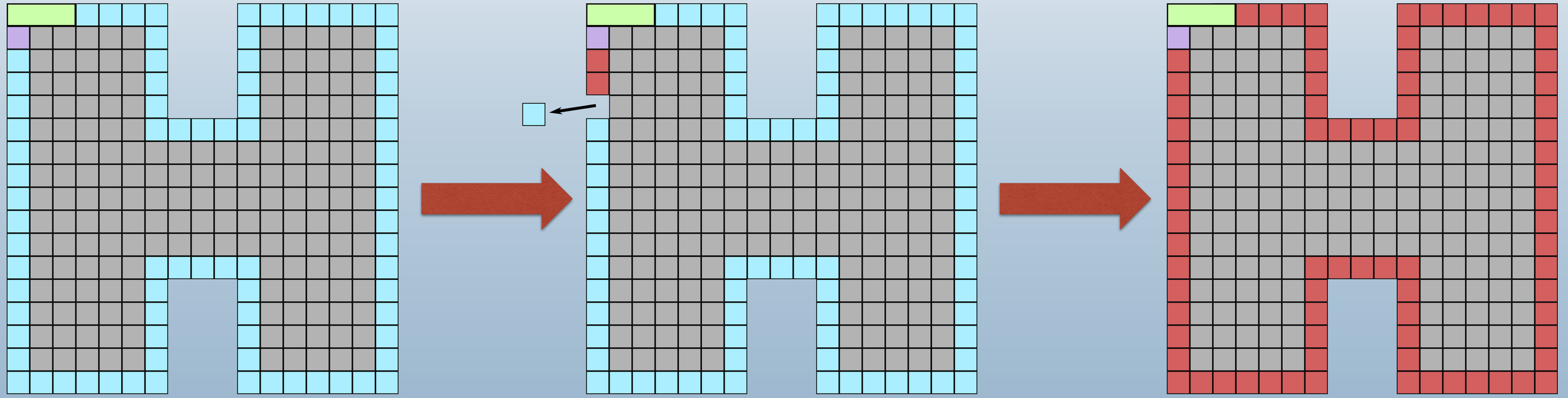
Theorem. *There exists a sleek universal shape replicator* $\Gamma = (\sigma, 10)$ *for genus-*0 *(hole-free) shapes with feature size of at least* 9.

High-level Overview



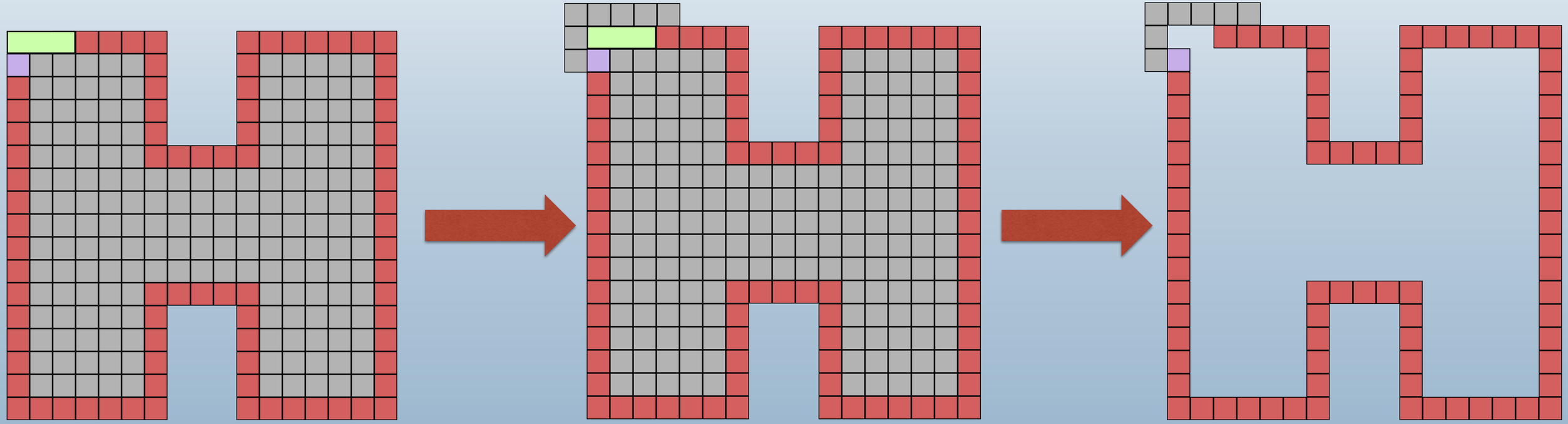
Phase 1

High-level Overview

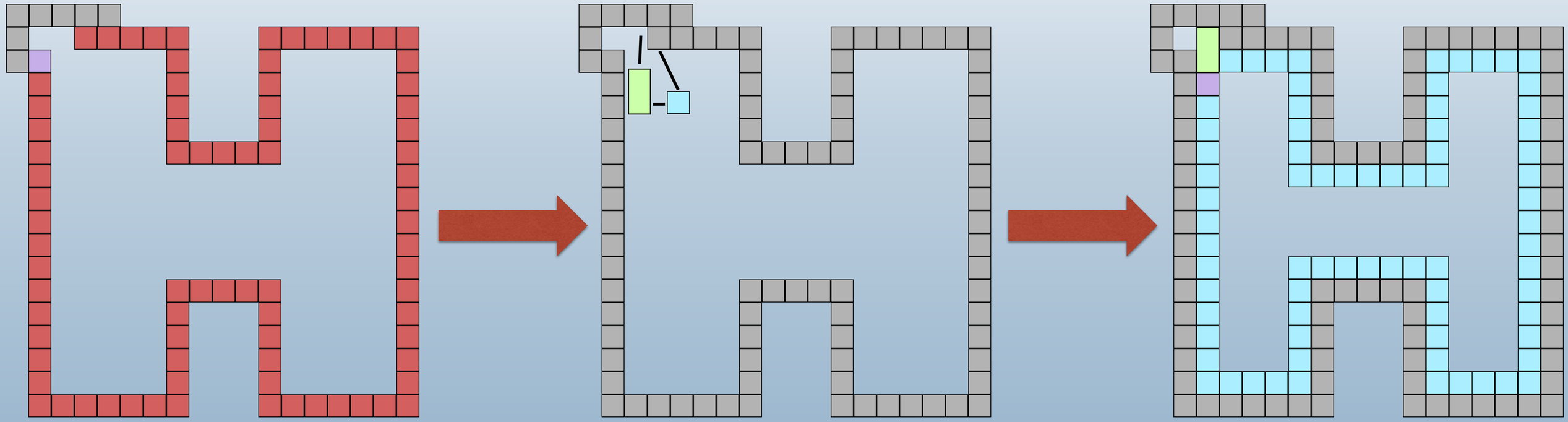


Phase 2

High-level Overview

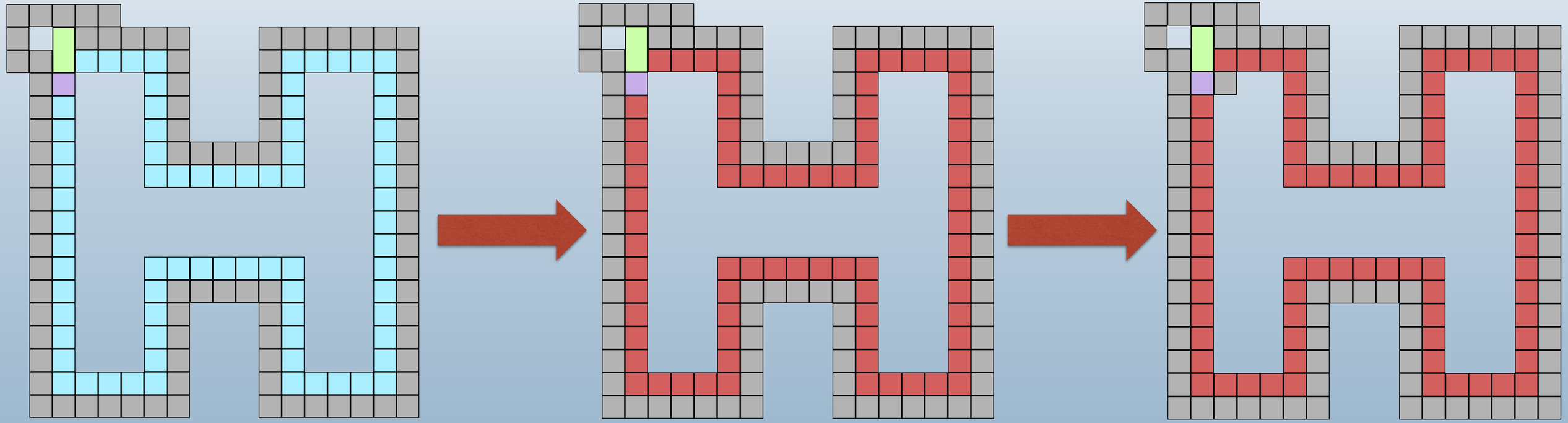


High-level Overview



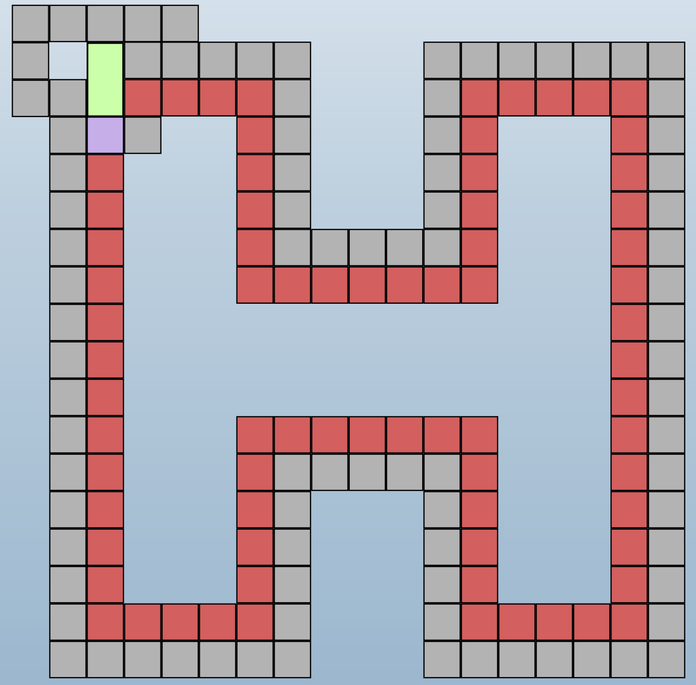
Phase 3

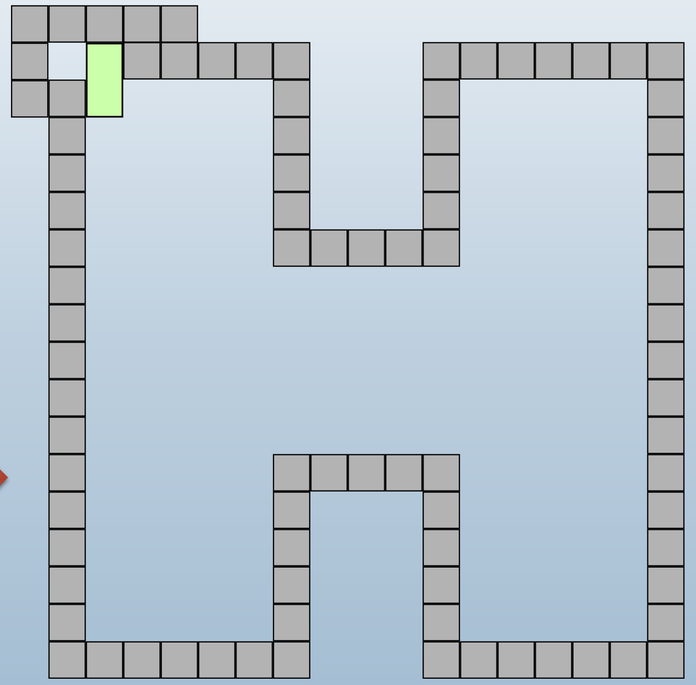
High-level Overview



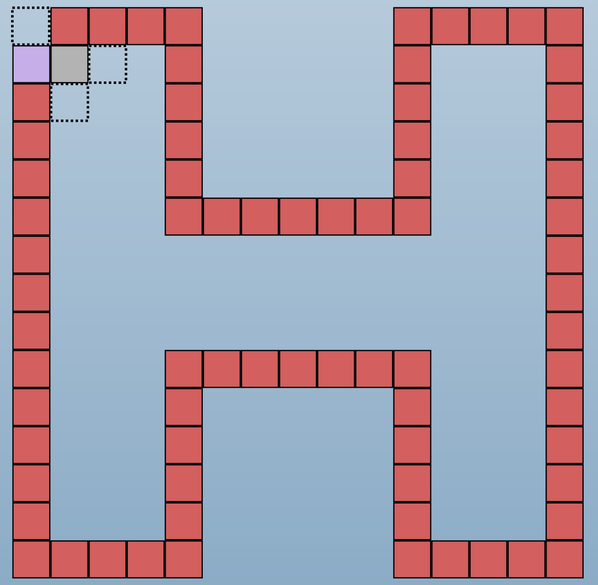
Phase 4

High-level Overview



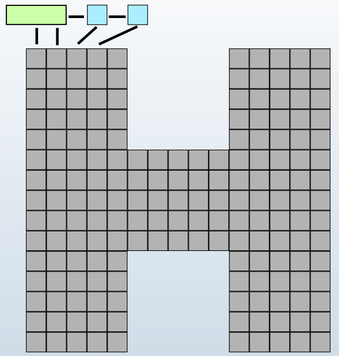


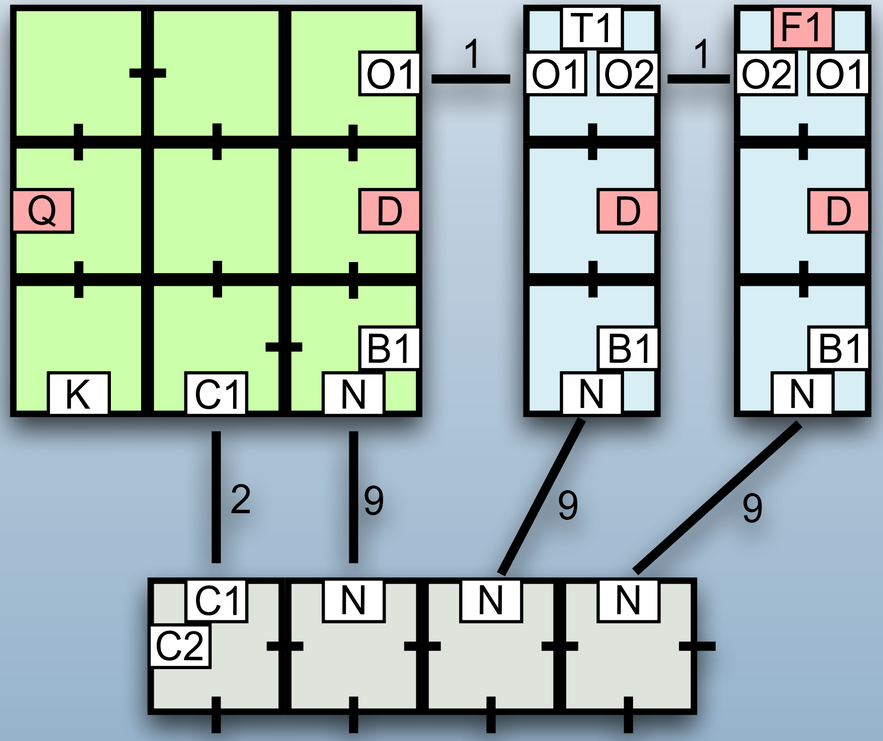
The frame can create another copy afterwards

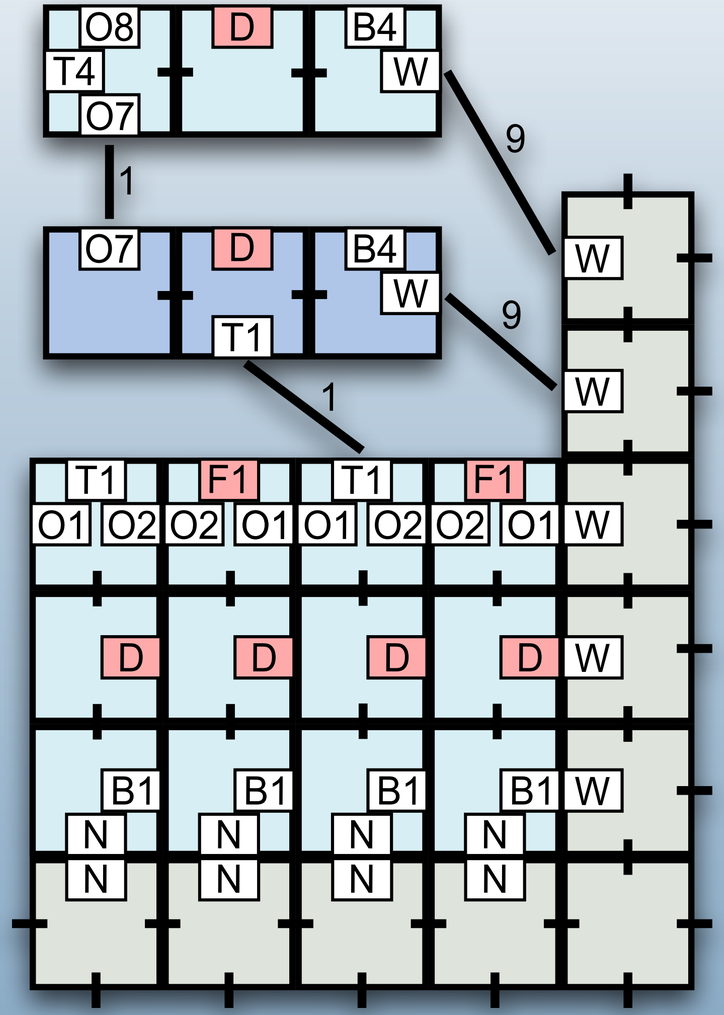


The shape fills and is a copy of the input shape

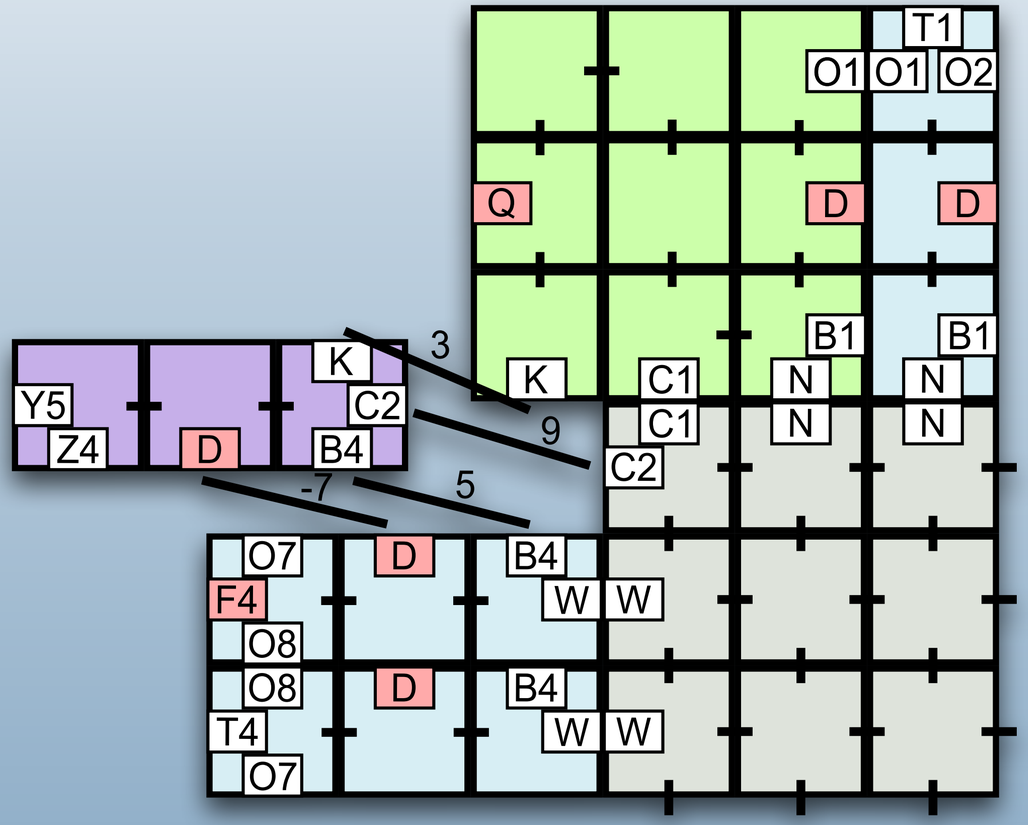
Some Details for Phase 1

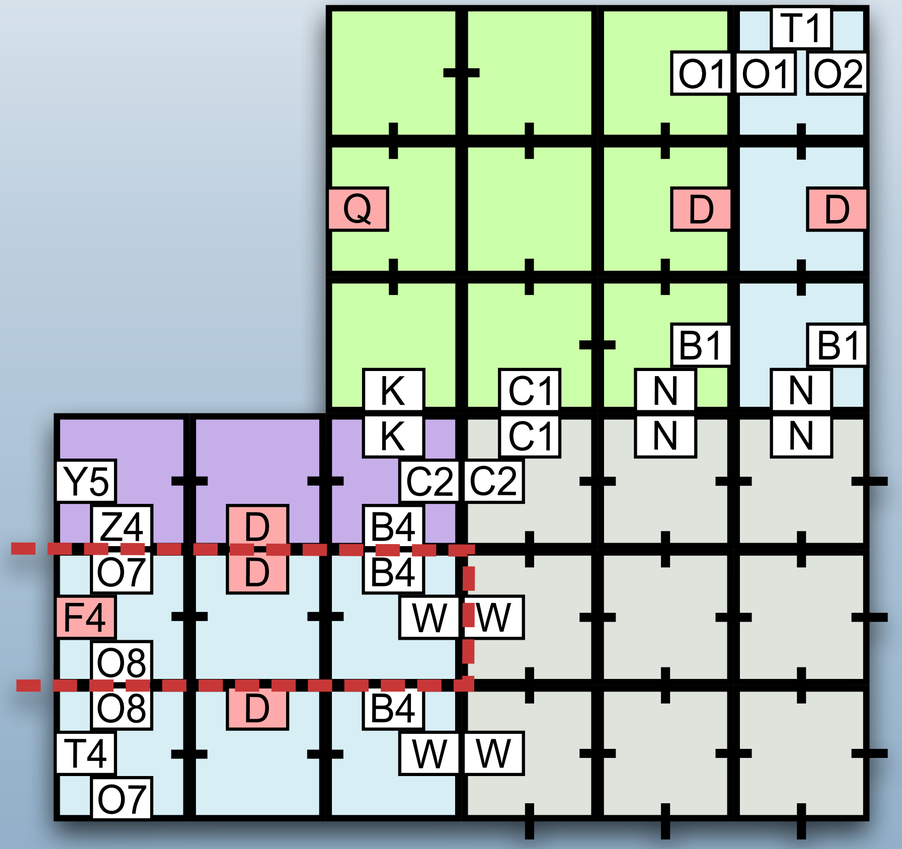


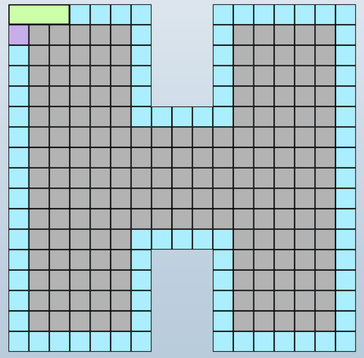




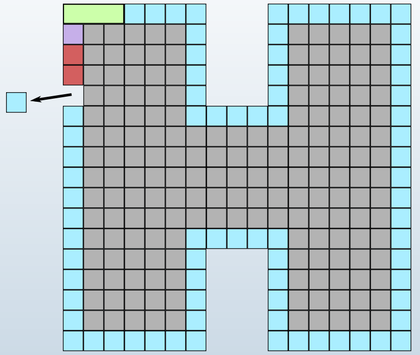
Some Details for Phase 1

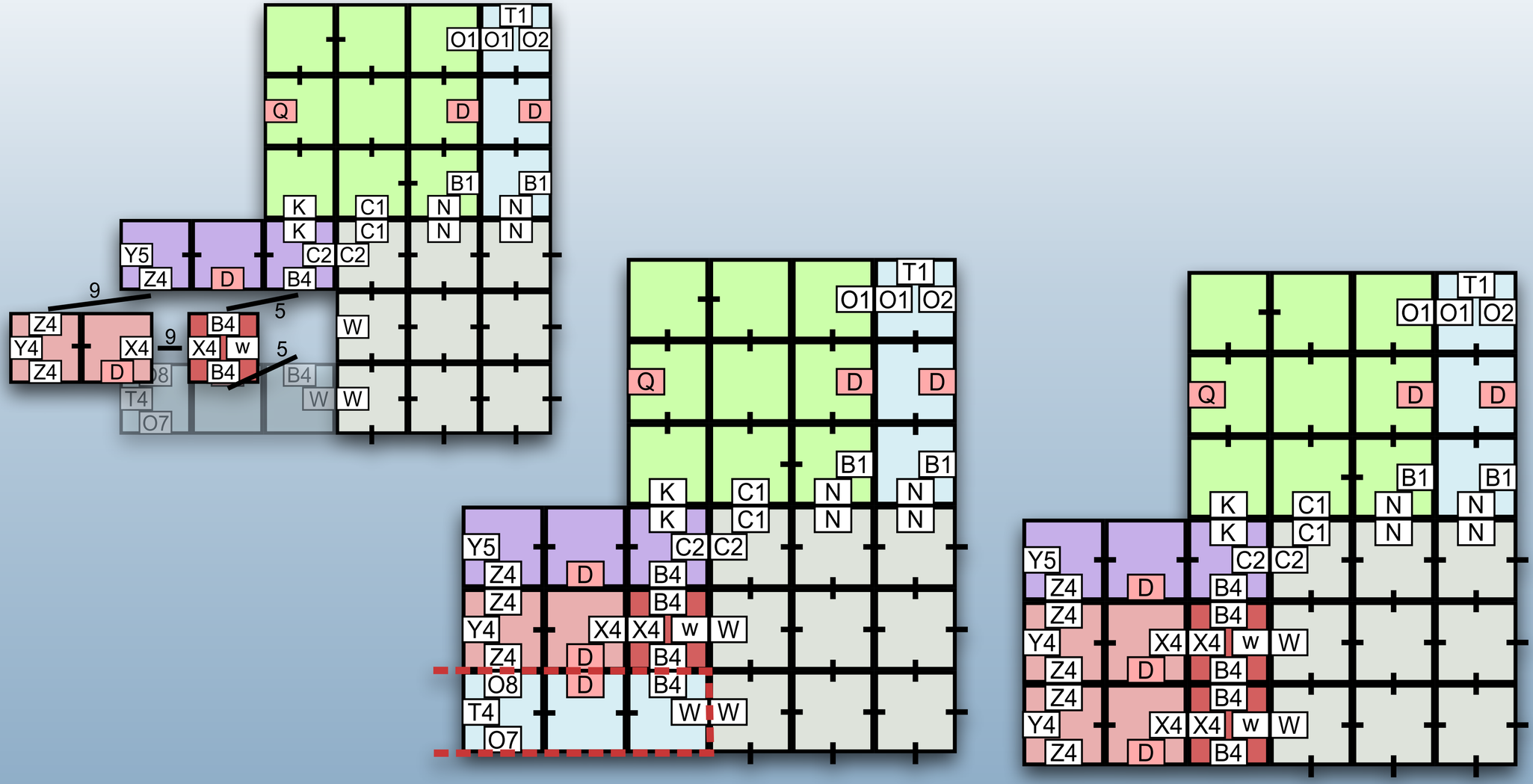




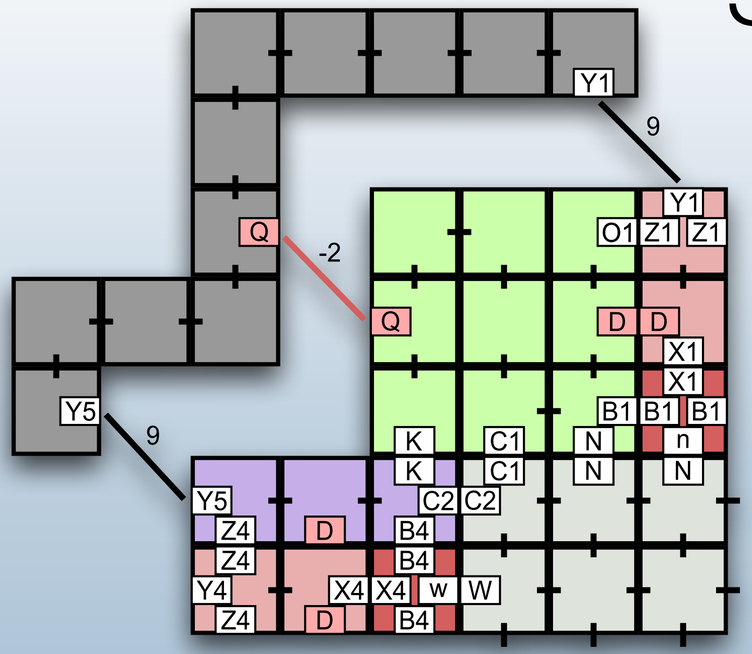


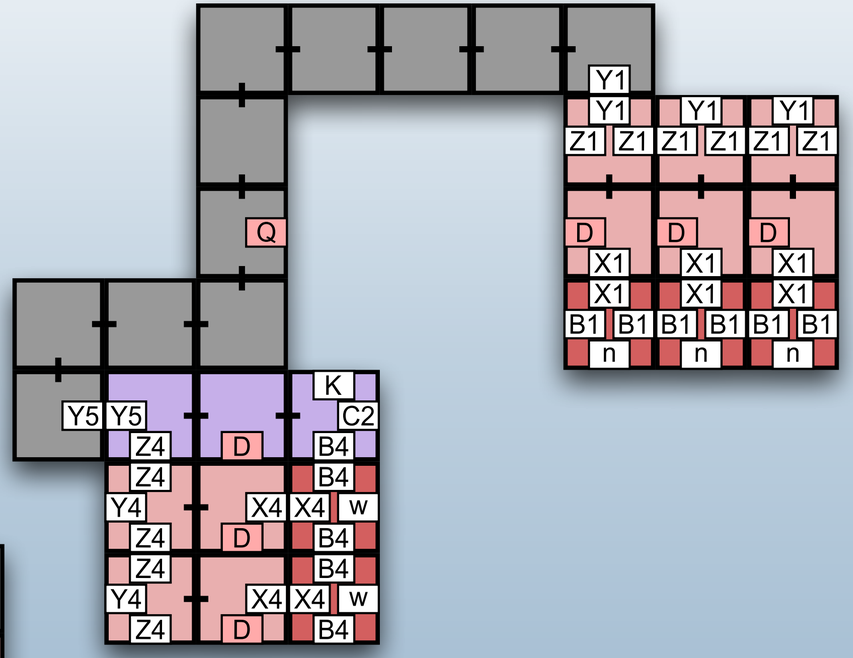
Some Details for Phase 2

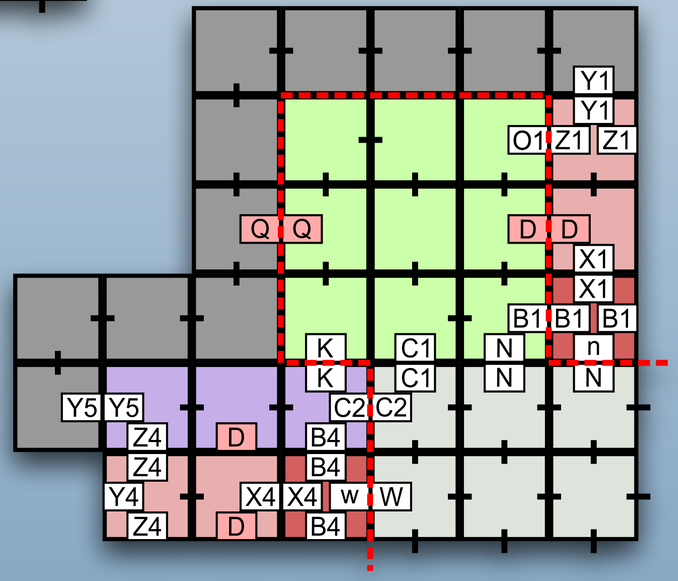


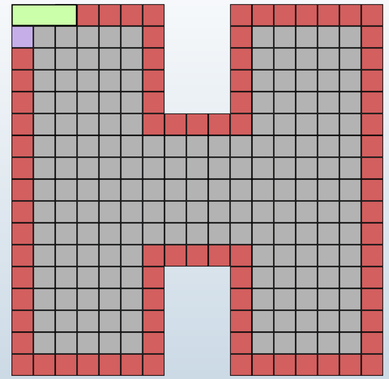


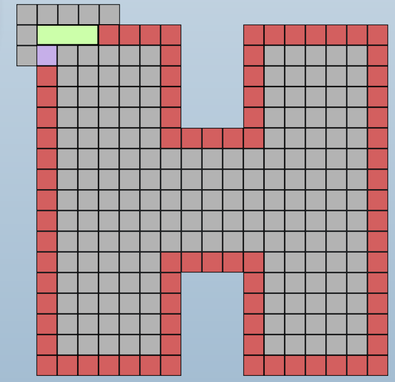
Some Details for Phase 2

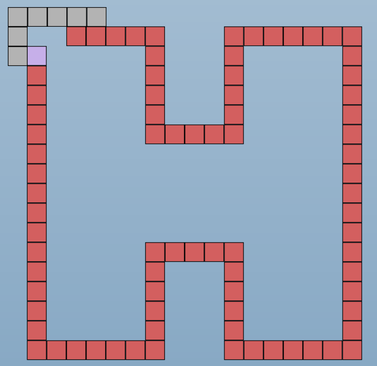






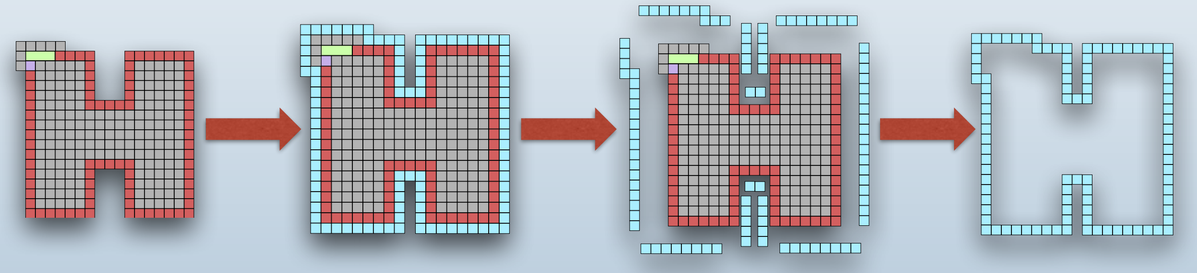






Future Work

Planar replication
- Enumerate the edges
- Break the frame apart
- Reassemble the pieces



Replication in 3D

Shapes with holes

Optimization of the current system
- Rate of replication
- Reduce the temperature

Constraints

Phase 1	
$C + N \geq \tau$	$O + N \geq \tau$
$T + W \geq \tau$	
Phase 2: Pre-drilling	
$K + C2 + B + D \geq \tau$	$K + C2 \geq \tau$
$B + W + O + D < \tau$	
Phase 2: Drilling	
$B + B \geq \tau$	$Z + X + D \geq \tau$
$B + X \geq \tau$	
Phase 2: Drilling Type-1 Corners	
$B + W + T + D < \tau$	$Y + T \geq \tau$
$Z + V + F \geq \tau$	$Y + Z + T + F \geq \tau$
$Z + B + V + Y + F \geq \tau$	$Z + X + V + Y + F \geq \tau$
$T + Z + B + F \geq \tau$	$O + N + O + F < \tau$
$B + B \geq \tau$	$B + U \geq \tau$
Phase 2: Drilling Type-2 Corners	
$Z + H + F \geq \tau$	$H + Y + Z + B + F \geq \tau$
$H + Y + Z + X + F \geq \tau$	$T + Z + B + F \geq \tau$
$O + N + O + F < \tau$	$N < \tau$
$A \geq \tau$	$B + B \geq \tau$
Phase 2: Drilling Obtuse Corners	
$B + N + O + D < \tau$	$B + B \geq \tau$
$Z + R + S + D \geq \tau$	$O + W + B + D < \tau$

Phase 2: Post-Drilling	
$B + N + C1 + K + D \geq \tau$	$Y + Y + Q \geq \tau$
$Y + Z + B + N + C1 + K + Q \geq \tau$	$B + N + C1 + C2 + B + Z + Y + Q + D \geq \tau$
$Y + Z + B + C2 + K + Q \geq \tau$	$B + N + C1 + K + D + Q \geq \tau$
$C2 + C1 + N \geq \tau$	$K + B + C2 + Q + D < \tau$
Phase 3	
$C2 + B \geq \tau$	$o + n \geq \tau$
$t + e \geq \tau$	
Phase 4 - Pre-drilling	
$k + w + b + D \geq \tau$	$b + w + i + D < \tau$
$k + w \geq \tau$	
Phase 4 - Drilling	
$b + b \geq \tau$	$z + x + D \geq \tau$
$b + w + i + D < \tau$	$b + x \geq \tau$
Phase 4 - Post-drilling	
$B + b + k + C2 + D \geq \tau$	$y + y + Q \geq \tau$
$B + b + k + C2 + D + Q \geq \tau$	$b + k + w + D + q < \tau$
$B + C2 \geq \tau$	$B + C2 + C1 + N + B1 + O + D \geq \tau$
$Z + B \geq \tau$	$B + W + W + O + D \geq \tau$
$k + y \geq \tau$	
Phase 5	
$L + g* \geq \tau$	$M + M \geq \tau$